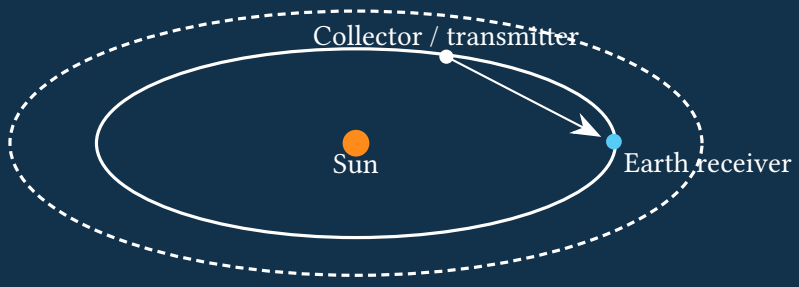
PROJECT DYSON SPHERE

The Mathematics Behind HelioSwarm

A mathematical textbook for implementation milestones M1–M6



Orbital dynamics → targeting → solar/thermal physics → diffraction → grid power

Source-grounded explanatory edition | 13 November 2026

Repository branch: `helioswarm-m0-m6` | inspected commit: `560675f`

Preface and scope

This volume explains the *implemented mathematics* in milestones M1–M6 of the `helioswarm` branch of *Project Dyson Sphere*. It is organized in physical dependency order: a satellite state must be well defined before it can be propagated; a propagated orbit and rotating terrestrial receiver are needed before a beam can be aimed; solar conversion and thermal constraints determine the electrical input; wave optics determines the intercepted fraction; and only then can the code calculate time-averaged grid power.

This is an **expository reconstruction**, not a verbatim copy of source files. The governing equations, parameter conventions, named approximations, test figures, and model distinctions have been checked against the repository's `docs/model_assumptions.md`, `docs/milestones/M1.md–M6.md`, code interfaces, and the original 28-page specification. Derivations and explanatory intermediate steps are supplied here to make the mathematics readable. Where the *plan* and the *implementation* differ, this textbook explicitly identifies the implemented choice. Statements called *validation* mean numerical or reference checks, not spacecraft hardware qualification.

How to use this textbook

M1 supplies the analytic two-body reference. **M2** computes realistic spacecraft motion. **M3** finds when and where Earth can receive a beam. **M4** computes electrical input and heat. **M5** computes the beam's finite-receiver capture fraction. **M6** combines the earlier components and produces actual grid-delivery and availability statistics. Each chapter has sequential equations, a local symbol guide, reasoning, assumptions, and testable consequences.

Scope limitation. No multi-satellite scheduling, Pareto optimization, GEO comparison, or experimental hardware demonstration is claimed here: those belong to M7–M9 or later work. "Every variable" in this volume refers to every symbol introduced in its presented mathematical equations, including symbols in the principal numerical methods; it is not an enumeration of every code field or implementation-local temporary variable.

Contents

Preface and scope i

Conventions, constants, and symbol grammar vi

1 M1 — Analytic orbital mechanics 1

1.1 The starting physical model: a solar two-body problem . . . 1

1.1.1 Angular momentum and the orbital plane . . . 1

1.1.2 Specific orbital energy . . . 2

1.2 Classical orbital elements and geometric meaning . . . 2

1.3 Mean motion and Kepler's third law . . . 3

1.4 Converting orbital elements to a Cartesian state . . . 4

1.4.1 Construct the orbital basis . . . 4

1.4.2 Recovering orbital elements from a state . . . 4

1.5 Why true, eccentric, and mean anomalies are all needed . . . 5

1.6 Solving Kepler's equation numerically . . . 5

1.7 Circular and equatorial orbit degeneracies . . . 6

1.8 Moving between fixed ecliptic and equatorial axes . . . 6

1.9 Step-by-step M1 propagation algorithm . . . 6

2 M2 — Ephemerides and orbital propagation 8

2.1 Why an ephemeris and time standard are necessary . . . 8

2.1.1 UTC to TDB, and what the symbols mean . . . 8

2.2 Geometric SPICE state vectors and origin shifts . . . 9

2.3 Gravitational acceleration: inertial and Sun-relative equations . . . 9

2.3.1 Derivation of the implemented heliocentric indirect term . . . 9

2.4 Deriving solar-radiation-pressure acceleration . . . 10

2.4.1 Exactly equivalent reduced-gravity parameter . . . 11

2.4.2 Secular along-track drift estimate . . . 11

2.5 Numerical first-order system and DOP853 . . . 11

2.6 Piecewise cubic Hermite ephemeris cache . . . 12

2.7 Independent validation and omitted forces . . . 12

3 M3 — Earth targeting, visibility, and occultation 14

3.1 Ground-site geodetic coordinates on WGS 84 . . . 14
3.2 Local up, east, and north unit vectors . . . 14
3.3 Earth rotation: an Earth-fixed station is not inertially fixed . . . 15
3.4 Finite light-time: target the receiver on arrival . . . 15
3.4.1 The implemented fixed-point iteration . . . 16
3.5 Relativistic aberration of transmitter and receiver directions . . . 16
3.6 Pointing-rate derivative and range rate . . . 17
3.7 Elevation and azimuth at the receiver . . . 17
3.8 Beam occultation as a finite-segment geometry problem . . . 17
3.8.1 Earth: ellipsoid rather than sphere . . . 18
3.8.2 Sun and Moon . . . 18
3.9 Solar eclipse and penumbra at the spacecraft . . . 18
3.10 Access fractions, aliasing, and interpretation . . . 19

4 M4 — Solar conversion and thermal equilibrium 20
4.1 Solar irradiance is an inverse-square flux . . . 20
4.2 Projected collection area and partial illumination . . . 20
4.3 Absorption, reflection, and what becomes heat . . . 21
4.4 Temperature-dependent photovoltaic conversion . . . 21
4.5 The PV thermal node: solve electricity and heat together . . . 21
4.5.1 Construct the nonlinear residual . . . 22
4.5.2 An analytically constructed solver bracket . . . 22
4.5.3 Uniqueness and float64 conditioning . . . 22
4.6 Electronics and beam-conversion thermal node . . . 23
4.7 Sizing the separate heat radiator . . . 23
4.8 Feasibility margins and optional radiator mass proxy . . . 24
4.9 The exact energy ledger: track every watt . . . 24
4.10 Worked solar/thermal comparison . . . 25

5 M5 — Diffraction, beam capture, pointing and atmosphere 26
5.1 What the electromagnetic field model assumes . . . 26
5.1.1 Separate the launch power from its spatial distribution . . . 26
5.2 Three implemented aperture-illumination profiles . . . 26
5.2.1 Uniform circular illumination . . . 27
5.2.2 Circularly truncated Gaussian illumination . . . 27
5.2.3 Parabolic taper on a nonzero pedestal . . . 27
5.3 Scalar diffraction from a circular aperture . . . 27

5.3.1 Far-field (Fraunhofer) limit . . . 28
5.3.2 The Fresnel number and model-validity decisions . . . 28
5.4 The Airy pattern: a uniformly illuminated circular aperture . . . 29
5.4.1 Encircled energy and the small-receiver limit . . . 29
5.5 Gaussian-beam alternative . . . 30
5.6 Off-axis receiver: exact finite-disk interception . . . 30
5.7 Receiver orientation and projected geometry . . . 31
5.8 Pointing bias and two-dimensional angular jitter . . . 31
5.8.1 Pointing efficiency and its caveat . . . 32
5.9 Wavefront phase-error reduction . . . 32
5.10 Microwave phased-array lattice factor . . . 32
5.11 Assemble beam capture without loss double-counting . . . 33
5.12 Atmospheric attenuation on the terminal Earth segment . . . 33
5.12.1 Simple atmospheric airmass approximation . . . 33
5.12.2 ITU-R microwave model: externally specified attenuation . . . 34
5.13 Physical scaling exercise: why interplanetary capture is difficult . . . 34

6 M6 — End-to-end single-satellite energy delivery 35
6.1 One satellite, one receiver, one time-indexed calculation . . . 35
6.2 Availability logic must be evaluated before the power budget . . . 35
6.3 The complete algebraic link budget . . . 36
6.3.1 Receiver heat-rejection constraint . . . 37
6.3.2 Maximum beam irradiance and beam-abort rule . . . 37
6.4 The power and energy identities . . . 38
6.5 Time integration: instantaneous watts versus annual megawatt-hours . . . 38
6.6 Availability, capacity factor and end-to-end efficiency . . . 38
6.6.1 What the capacity-factor denominator actually means . . . 39
6.6.2 End-to-end incident-sunlight efficiency . . . 39
6.7 Outage and reason statistics . . . 39
6.7.1 Sampling diagnostics . . . 40
6.8 Worked baseline: what the current simulator actually predicts . . . 40
6.8.1 Why a much larger optical aperture can improve the result . . . 40
6.9 Reproducible outputs and what each proves . . . 41
6.10 Numerical performance and the optional JAX kernel . . . 41
6.11 M6 research interpretation and what remains outside its scope . . . 41

A Cross-milestone equation map and symbol index 42

A.1 Sequential dependency map 42
A.2 Frequently confused variables and names 42
A.3 Validation checklist by mathematical requirement 43

B Sources and reproducibility 44
B.1 Repository snapshot and interpretation policy 44
B.1.1 Primary repository documentation 44
B.2 Key theoretical and standards sources named in the repository 44
B.3 What can and cannot be concluded from the implemented mathematics 45

Conventions, constants, and symbol grammar

Unit system. All computational quantities are in SI: metre (m), second (s), kilogram (kg), watt (W), kelvin (K), radian (rad). Configuration files alone use au, degrees, days, gigahertz, and decibels where explicit in field names. An angle in a trigonometric function is in radians unless stated otherwise. Bold symbols are Cartesian vectors, unbold symbols scalars; $\hat{\mathbf{u}}$ denotes a unit vector, $\mathbf{a} \cdot \mathbf{b}$ a scalar product, and $\|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}}$ a Euclidean norm.

Symbol	Value / unit		Definition and provenance
au	149 597 870 700 m exact		Astronomical unit (IAU 2012 B2).
c	299 792 458 m s^{-1} exact		Vacuum speed of light, SI definition.
$\mu_\odot = GM_\odot$	1.3271244004127942 $\times$ 10^{20} m^3 s^{-2}		DE440-compatible solar gravitational parameter in kernel-free M1; SPICE supplies kernel value in M2.
σ_{SB}	5.670374419 $\times$ 10^{-8} W m^{-2} K^{-4}		Stefan–Boltzmann constant (CODATA 2018).
$S_{1\mathrm{au}}$	1361 W m^{-2} default		Nominal solar irradiance at 1 au; configurable in the code.
$R_\odot$	6.957×10^8 m		Nominal solar radius (IAU 2015 B3).
$a_\oplus$	6 378 137 m		WGS-84 Earth equatorial semimajor axis.
$f_\oplus$	1/298.257223563		WGS-84 Earth flattening.
R_{Moon}	1 737 400 m		Mean Moon radius used for occultation.
1 day	86 400 s exact		Uniform SI-day conversion.
ε_0	84 381.448 arcsec		J2000 mean obliquity used by SPICE's ECLIPJ2000 frame.
ϵ_{mach}	$\approx 2.22 \times 10^{-16}$		Float64 machine epsilon; not the physical emissivity ϵ.

Time and coordinates. A subscript $\odot$ denotes the Sun, $\oplus$ Earth, and j another gravitating body; t is a dynamical epoch in TDB seconds relative to J2000. All combined states are expressed in J2000 Cartesian axes and normally referred to the Solar System Barycenter (SSB). Sun-centred classical elements are configured in the fixed ECLIPJ2000 plane. Angles Ω and ω denote node and periapsis angles; the same Greek letter in another chapter must be distinguished by its subscript and local definition.

Notation discipline. The letters a, e, i, D, R, T, A, η, ϵ, and β are reused in different physical models; each is redefined at first use in its chapter. In particular, orbital eccentricity e is not Earth flattening $f_\oplus$; optical aperture radius a_{ap} is not orbital semimajor axis a; and β_{SRP} (solar-radiation-pressure lightness) is not $\boldsymbol{\beta} = \mathbf{v}/c$ (relativistic speed).

M1 — Analytic orbital mechanics

Primary implementation source: `constants.py`, `frames.py`, `orbit.py`; `docs/milestones/M1.md`; `docs/model_assumptions.md`, M1.

Key idea

M1 creates a mathematically exact (up to floating-point round-off) Keplerian reference orbit. Its purpose is not to predict every real perturbation, but to establish the coordinates, invariants, transformations and solver against which M2's full propagator can be verified.

1.1 The starting physical model: a solar two-body problem

Assume a spacecraft whose mass is negligible compared with the Sun. Ignore all planets, solar-radiation pressure, the solar reflex motion, relativity, thrust and collisions. Treat the Sun as a point mass and the spacecraft as a point test body. Newton's inverse-square force gives the *relative* equation

$$\ddot{\mathbf{r}} = -\frac{\mu_\odot}{r^3}\mathbf{r}, \qquad r = \|\mathbf{r}\|, \quad \mu_\odot = GM_\odot. \tag{1.1}$$

Definitions. $\mathbf{r}(t)$ is the Sun-to-spacecraft displacement [m]; $\ddot{\mathbf{r}} = \mathrm{d}^2\mathbf{r}/\mathrm{d}t^2$ is relative acceleration [m s^{-2}]; r is distance [m]; $\mu_\odot$ is the solar gravitational parameter [m^3 s^{-2}]; G is Newton's gravitational constant; and $M_\odot$ is solar mass. The mass of the test particle cancels from acceleration.

The $r^{-3}\mathbf{r}$ expression is a vector way of writing an inward acceleration of magnitude $\mu_\odot/r^2$. Because the acceleration is central, it exerts zero torque about the Sun and permits exact orbital invariants.

1.1.1 Angular momentum and the orbital plane

Take the cross product of $\mathbf{r}$ with Eq. (1.1). Define the *specific angular momentum*

$$\mathbf{h} = \mathbf{r} \times \mathbf{v}, \qquad \frac{\mathrm{d}\mathbf{h}}{\mathrm{d}t} = \mathbf{r} \times \ddot{\mathbf{r}} = \mathbf{0}, \qquad \mathbf{v} = \dot{\mathbf{r}}. \tag{1.2}$$

Definitions. $\mathbf{h}$ is angular momentum per unit spacecraft mass [m^2 s^{-1}]; $\mathbf{v}$ is relative velocity [m s^{-1}]; and $\times$ denotes the cross product.

Thus $\mathbf{h}$ is a constant vector normal to the fixed orbital plane. A normalized normal is $\hat{\mathbf{h}} = \mathbf{h}/\|\mathbf{h}\|$. A radial gravitational force cannot change this direction. Consequently, the inclination and longitude of ascending node are constant in M1.

1.1.2 Specific orbital energy

Dot Eq. (1.1) with velocity and integrate:

$$\mathbf{v} \cdot \ddot{\mathbf{r}} = -\mu_\odot \frac{\mathbf{v} \cdot \mathbf{r}}{r^3}, \tag{1.3}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{v^2}{2} - \frac{\mu_\odot}{r}\right) = 0, \qquad \mathscr{E} = \frac{v^2}{2} - \frac{\mu_\odot}{r}. \tag{1.4}$$

Definitions. $v = \|\mathbf{v}\|$ is speed; $\mathscr{E}$ is specific mechanical energy [$\mathrm{m^2\ s^{-2}}$], with kinetic term $v^2/2$ and gravitational potential $-\mu_\odot/r$.

For a bound ellipse $\mathscr{E} < 0$, and its semimajor axis satisfies

$$\mathscr{E} = -\frac{\mu_\odot}{2a}, \qquad v^2 = \mu_\odot\left(\frac{2}{r} - \frac{1}{a}\right). \tag{1.5}$$

Definitions. $a > 0$ is the orbital semimajor axis [m]; the second identity is the vis-viva equation.

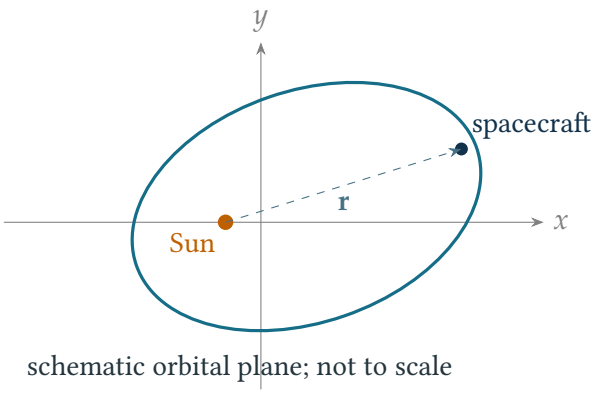
Why introduce the energy law? It gives two independent tests: (i) the state should map back to its original a, and (ii) $\mathscr{E}$ must be constant in M1, unlike M2 when planets add a time-dependent perturbation.

1.2 Classical orbital elements and geometric meaning

An elliptic Kepler orbit is represented by six osculating elements

$$(a, e, i, \Omega, \omega, \nu), \tag{1.6}$$

where a is semimajor axis [m], e is eccentricity [dimensionless, $0 \leqslant e < 1$], i is inclination [rad] to the reference x–y plane, Ω is longitude of the ascending node [rad], ω is argument of periapsis [rad], and ν is true anomaly [rad], measured from periapsis to current position in the direction of motion.



schematic orbital plane; not to scale

The *semilatus rectum* p and radial orbit equation are

$$p = a(1 - e^2) = \frac{h^2}{\mu_\odot}, \qquad r(\nu) = \frac{p}{1 + e\cos\nu}, \qquad h = \|\mathbf{h}\|. \tag{1.7}$$

Where the conic equation comes from

Central-force motion has specific angular momentum $h = r^2\dot{\varphi}$, with φ the polar angle within the orbital plane. Set $u_r(\varphi) = 1/r(\varphi)$, the reciprocal radial distance. Rewriting radial acceleration using φ as the

independent variable gives the Binet equation

$$\frac{\mathrm{d}^2 u_r}{\mathrm{d}\varphi^2} + u_r = \frac{\mu_\odot}{h^2}. \tag{1.8}$$

Definitions. u_r has units m^{-1} and is not the argument of latitude u used later; φ is the in-plane polar angle [rad], $\dot{\varphi}$ its time derivative [rad/s], and h is specific angular momentum.

The constant-coefficient solution is

$$u_r(\varphi) = \frac{\mu_\odot}{h^2}\left[1 + e\cos(\varphi - \omega_p)\right]. \tag{1.9}$$

Definitions. ω_p is the polar angle of periapsis within the chosen orbital plane [rad]; e is the dimensionless integration constant describing eccentricity.

With $\nu = \varphi - \omega_p$ and $p = h^2/\mu_\odot$, inversion recovers $r = p/(1 + e\cos\nu)$. Hence the conic is a consequence of the inverse-square central force, not an arbitrary empirical curve.

The two special distances follow immediately:

$$r_{\mathrm{peri}} = a(1 - e), \qquad r_{\mathrm{apo}} = a(1 + e). \tag{1.10}$$

Definitions. p is semilatus rectum [m]; h is specific angular-momentum magnitude [$\mathrm{m}^2\ \mathrm{s}^{-1}$]; r_{peri} and r_{apo} are perihelion and aphelion distances [m].

At $\nu = 0$, $\cos\nu = 1$ and the spacecraft is nearest the Sun; at $\nu = \pi$, $\cos\nu = -1$ and it is farthest. M1 accepts only elliptic orbits and rejects configured perihelia inside the nominal solar radius $R_\odot$.

1.3 Mean motion and Kepler's third law

Define the *mean motion*, the average anomaly-advance rate, by

$$n = \sqrt{\frac{\mu_\odot}{a^3}}, \qquad T_{\mathrm{orb}} = \frac{2\pi}{n} = 2\pi\sqrt{\frac{a^3}{\mu_\odot}}. \tag{1.11}$$

Definitions. n [rad s^{-1}] is mean motion, and T_{orb} [s] is the full orbital period; π is the usual circle constant.

One can derive this scaling geometrically. The ellipse has semiminor axis $b = a\sqrt{1 - e^2}$ and area πab. Kepler's equal-area law follows from conserved h: the areal speed is $\mathrm{d}\mathscr{A}/\mathrm{d}t = h/2$. Therefore

$$T_{\mathrm{orb}} = \frac{\pi ab}{h/2} = \frac{2\pi a^2\sqrt{1 - e^2}}{\sqrt{\mu_\odot a(1 - e^2)}} = 2\pi\sqrt{\frac{a^3}{\mu_\odot}}. \tag{1.12}$$

Definitions. b is ellipse semiminor axis [m] and $\mathscr{A}$ is swept orbital area [m^2], unrelated to collector area. The relation $h^2 = \mu_\odot a(1 - e^2)$ came from Eq. (1.7).

Thus $n^2 a^3 = \mu_\odot$ for all elliptic eccentricities. Therefore, increasing a increases the period according to $T_{\mathrm{orb}} \propto a^{3/2}$. *Eccentricity changes how fast the spacecraft travels at each point, not its period for fixed a.*

1.4 Converting orbital elements to a Cartesian state

Numerical propagation, Earth targeting, and SPICE use position/velocity vectors rather than six elements. The repository therefore implements the element-to-state map explicitly.

1.4.1 Construct the orbital basis

Define two orthonormal vectors within the orbital plane,

$$\mathbf{P} = (\cos\Omega, \sin\Omega, 0), \tag{1.13}$$

$$\mathbf{Q} = (-\sin\Omega\cos i, \cos\Omega\cos i, \sin i), \tag{1.14}$$

$$u = \omega + \nu. \tag{1.15}$$

Definitions. $\mathbf{P}$ points along the ascending-node direction [dimensionless]; $\mathbf{Q}$ is 90 degrees ahead along the inclined orbital plane [dimensionless]; u is argument of latitude [rad], the current in-plane angle measured from the node.

Here the Greek letter in ν above denotes true anomaly; u is their sum, not an independent orbital element. The preceding line is to be read as $u = \omega + \nu$.

The state vectors are

$$\mathbf{r} = r\,(\cos u\,\mathbf{P} + \sin u\,\mathbf{Q})\,, \tag{1.16}$$

$$\mathbf{v} = \sqrt{\frac{\mu_\odot}{p}}\,[-(\sin u + e\sin\omega)\mathbf{P} + (\cos u + e\cos\omega)\mathbf{Q}]\,. \tag{1.17}$$

All terms in the position bracket are dimensionless; multiplying by r yields metres. The prefactor $\sqrt{\mu_\odot/p}$ in the velocity equation has units m s^{-1}. These equations equal the textbook perifocal-state formula rotated successively by the node, inclination, and periapsis angles. In code, they should agree with the independent three-rotation construction to floating-point precision.

1.4.2 Recovering orbital elements from a state

Given a nondegenerate Sun-centred state $(\mathbf{r}, \mathbf{v})$, compute

$$\mathbf{h} = \mathbf{r}\times\mathbf{v}, \qquad \mathbf{e} = \frac{\mathbf{v}\times\mathbf{h}}{\mu_\odot} - \frac{\mathbf{r}}{r}, \qquad e = \|\mathbf{e}\|\,, \tag{1.18}$$

$$\mathscr{E} = \frac{v^2}{2} - \frac{\mu_\odot}{r}, \qquad a = -\frac{\mu_\odot}{2\mathscr{E}}. \tag{1.19}$$

Definitions. $\mathbf{e}$ is the dimensionless eccentricity vector, pointing toward periapsis, and e is its magnitude.

The reference-plane inclination and node angle can then be recovered as

$$i = \operatorname{atan2}\left(\sqrt{h_x^2 + h_y^2}, h_z\right), \tag{1.20}$$

$$\Omega = \operatorname{atan2}(h_x, -h_y), \tag{1.21}$$

$$\omega = \operatorname{atan2}(\mathbf{e}\cdot\mathbf{Q}, \mathbf{e}\cdot\mathbf{P}), \tag{1.22}$$

$$u = \operatorname{atan2}(\mathbf{r}\cdot\mathbf{Q}, \mathbf{r}\cdot\mathbf{P}) - \omega. \tag{1.23}$$

Definitions. h_x, h_y, h_z are Cartesian components of $\mathbf{h}$; $\operatorname{atan2}(y, x)$ is the quadrant-aware angle function. Ω,

ω, and ν must be wrapped consistently into the chosen angular interval.

The orbit is rejected if its energy is nonnegative, because this implementation supports ellipses rather than unbound hyperbolic trajectories.

1.5 Why true, eccentric, and mean anomalies are all needed

The true anomaly ν is geometrically intuitive but does not increase uniformly in time. Define *eccentric anomaly* E and *mean anomaly* M by

$$M = E - e \sin E, \qquad M(t) = M_0 + n(t - t_0). \tag{1.24}$$

Definitions. E and M are angles [rad]; M_0 is the mean anomaly at reference epoch t_0 [TDB s]; t is the requested epoch [TDB s].

The linear equation for $M(t)$ is the reason the analytic propagator is so efficient: orbital motion is first advanced in a uniform-in-time variable, and then geometrically mapped back to ν.

The implementation uses continuous anomaly relations

$$\beta_e = \frac{e}{1 + \sqrt{1 - e^2}}, \tag{1.25}$$

$$\nu = E + 2 \operatorname{atan2}\Big(\beta_e \sin E, 1 - \beta_e \cos E\Big), \tag{1.26}$$

$$E = \nu - 2 \operatorname{atan2}\Big(\beta_e \sin \nu, 1 + \beta_e \cos \nu\Big). \tag{1.27}$$

Definitions. β_e is a dimensionless eccentricity transform (*not* SRP lightness or relativistic speed). The anomaly values are interpreted consistently across complete revolutions.

The familiar half-angle identity $\tan(\nu/2) = \sqrt{(1+e)/(1-e)} \tan(E/2)$ also holds, but a naive arctangent implementation can select the wrong quadrant or lose revolution continuity. The `atan2` forms avoid that ambiguity.

1.6 Solving Kepler's equation numerically

After advancing $M(t)$, Eq. (1.24) must be inverted for E. Let

$$f(E) = E - e \sin E - M, \quad f'(E) = 1 - e \cos E, \quad E_{k+1} = E_k - \frac{f(E_k)}{f'(E_k)}. \tag{1.28}$$

Definitions. f is the dimensionless Kepler residual [rad], f' its derivative [dimensionless], k the Newton iteration index, and E_k the current estimate [rad].

For an ellipse, $f'(E) > 0$ because $0 \leqslant e < 1$. Thus the inverse is unique for any unwrapped M. The code first reduces the anomaly into a symmetric interval, brackets the positive root by $[x, \min(x+e, \pi)]$ for $x \in [0, \pi]$, starts near $x + 0.85e$, and clips Newton iterates into the shrinking bracket.

A solver stops when its correction is sufficiently small ($\sim 10^{-12}$ rad), or when the residual reaches a backward-error floor set by floating-point round-off. The second criterion matters near $e \to 1$: $f' = 1 - e \cos E$ can be very small, so a tiny residual does not automatically imply a tiny forward error $\Delta E \approx f/f'$. An iteration cap raises a diagnostic instead of silently returning an unconverged orbit.

1.7 Circular and equatorial orbit degeneracies

When $e = 0$, there is no unique periapsis direction; ω is undefined. When $i = 0$ or $i = \pi$, the orbital plane does not have a uniquely defined ascending node; Ω is undefined. The repository fixes a canonical representation for numerical stability:

- For $e < 10^{-13}$, set $e = 0$, $\omega = 0$, and use argument of latitude as ν.
- For $\sin i < 10^{-13}$, set $\Omega = 0$ with node line along $+x$.
- For both circular and equatorial orbits, use true longitude for the remaining in-plane angle.

These are not new physical assumptions about a particular orbit. They are coordinate conventions preventing undefined angular parameters from contaminating state conversions.

1.8 Moving between fixed ecliptic and equatorial axes

Configured elements use ECLIPJ2000, while M2 uses SPICE's J2000 frame. With ε_0 the fixed mean obliquity, the implemented coordinate mapping is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{\mathrm{J2000}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\varepsilon_0 & -\sin\varepsilon_0 \\ 0 & \sin\varepsilon_0 & \cos\varepsilon_0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{\mathrm{ECLIPJ2000}} . \tag{1.29}$$

Definitions. x, y, z are position or velocity components in the indicated Cartesian frame; ε_0 is the fixed J2000 obliquity. The same matrix acts on $\mathbf{r}$ and $\mathbf{v}$ because both frames are inertial.

Rotating axes is *not* the same as changing the origin. Translating from Sun-centred to barycentric states requires the Sun's ephemeris, which begins in M2.

1.9 Step-by-step M1 propagation algorithm

1. Parse valid $(a, e, i, \Omega, \omega, \nu)$ and a TDB reference epoch t_0; reject $a \leqslant 0$, $e \notin [0, 1)$, or a perihelion inside the Sun.
2. Convert the reference true anomaly ν_0 to E_0, then $M_0 = E_0 - e \sin E_0$.
3. Compute $n = \sqrt{\mu_\odot / a^3}$ once.
4. For each epoch t, evaluate $M(t) = M_0 + n(t - t_0)$.
5. Solve $E - e \sin E = M(t)$ to the specified round-off-aware tolerance.
6. Convert E to ν, obtain $r = p/(1 + e\cos\nu)$, and evaluate Eq. (1.16).
7. Preserve the frame, origin, time scale and epoch metadata in the returned state.

Verification / interpretation

M1 acceptance tests. A circular orbit must retain radius and specific energy, a propagated orbit must return after T_{orb}, Cartesian-to-elements-to-Cartesian round trips must reconstruct the original state, Kepler's solver must remain stable for high eccentricities, and trajectories must agree with an independent numerical integration. The repository reports these comparisons in `docs/validation_report.md`, M1.

Assumptions and limits

Domain of validity. Exactly Keplerian only: one point-mass Sun and one massless satellite, no planets or SRP. The orbital plane and elements are fixed, and the solution has no numerical time-integration truncation error. Kepler's analytic propagator must not be interpreted as a precision heliocentric trajectory for a real solar sail or giant collector over long durations; that is why M2 exists.

M2 — Ephemerides and orbital propagation

Primary implementation source: `kernels.py`, `timescale.py`, `ephemeris.py`, `srp.py`, `propagate.py`; `docs/milestones/M2.md`; `docs/model_assumptions.md`, M2.

Key idea

M2 replaces the isolated two-body test orbit with a numerical massless spacecraft that feels prescribed DE440 solar/planetary gravity and configurable radial solar-radiation pressure. The central implementation choice is a Sun-relative equation with an *indirect acceleration* consistent with the very same included planetary bodies.

2.1 Why an ephemeris and time standard are necessary

M1 knows an abstract Kepler orbit around a fixed point-mass Sun. Real planets and the Sun move in an inertial barycentric frame, and any Earth-facing transmitter must be able to combine spacecraft and planetary states at the *same physical epoch*. The implementation uses four NAIF SPICE kernels: `naif0012.tls` for leap seconds, `pck00011.tpc` for body rotation and radii, `gm_de440.tpc` for gravitational parameters, and `de440s.bsp` for planetary states. The exact file hashes are retained for reproducibility.

2.1.1 UTC to TDB, and what the symbols mean

A text timestamp in UTC is converted through SPICE into an ephemeris time t_{TDB}: seconds of barycentric dynamical time past J2000. Schematically, the SPICE kernel time model takes the form

$$\mathrm{TDB} = \mathrm{UTC} + \Delta AT + 32.184\,\mathrm{s} + K \sin(M_{\mathrm{time}} + E_B \sin M_{\mathrm{time}}), \tag{2.1}$$

Definitions. ΔAT [s] is the cumulative TAI–UTC leap-second offset from the leap-second kernel; 32.184 s is TT–TAI; $K = 1.657 \times 10^{-3}$ s is a small periodic coefficient; M_{time} [rad] is the time-model mean-anomaly argument; E_B [dimensionless] is a kernel-defined correction in that periodic argument. The full conversion is performed by SPICE rather than recomputed using this schematic equation.

TDB and UTC are not interchangeable labels on the same number. For example, a one-minute timestamp error moves the Earth by thousands of kilometres along its orbit. The program never accepts an unlabeled UTC string as though it were already TDB.

A time grid is constructed as

$$t_k = t_0 + k\Delta t, \qquad k = 0, 1, \ldots, N, \qquad T = N\Delta t, \tag{2.2}$$

Definitions. t_0 is the initial TDB epoch [s], Δt is configured output cadence [s], k is a nonnegative integer, N is the number of intervals, and T is simulated duration [s].

The grid is uniform in TDB, not necessarily UTC. The implementation requires T to be a sufficiently precise integer multiple of Δt so that the final epoch equals the requested end of the window.

2.2 Geometric SPICE state vectors and origin shifts

The ephemeris provides the geometric barycentric state for each body j:

$$\mathbf{x}_j(t) \quad [\,\mathrm{m}], \qquad \mathbf{v}_j(t) = \frac{\mathrm{d}\mathbf{x}_j}{\mathrm{d}t} \quad [\,\mathrm{m\,s^{-1}}]. \tag{2.3}$$

Definitions. $\mathbf{x}_j$ and $\mathbf{v}_j$ are Cartesian position and velocity in J2000 axes with origin SSB; j may denote the Sun, a planet, a system barycenter, Earth, or Moon. SPICE's geometric vectors are returned in km and km/s and converted to SI using 10^3.

Planetary states are *geometric*, with no built-in light-time correction, since M3 explicitly computes propagation to the receiver. This avoids accidentally applying the delay twice.

If $\mathbf{r}_{s/\odot}$ is a spacecraft state relative to the Sun, the equivalent barycentric state is

$$\mathbf{x}_s = \mathbf{x}_\odot + \mathbf{r}_{s/\odot}, \qquad \mathbf{v}_s = \mathbf{v}_\odot + \dot{\mathbf{r}}_{s/\odot}. \tag{2.4}$$

Definitions. s denotes satellite, $\mathbf{x}_\odot$ is Sun position relative to SSB, and $\mathbf{r}_{s/\odot} = \mathbf{x}_s - \mathbf{x}_\odot$. All vectors must refer to the same epoch and axes before addition/subtraction.

To initialize the spacecraft, convert M1's Sun-centred ECLIPJ2000 elements into Cartesian state, rotate into J2000 using M1's obliquity matrix, and add the Sun's J2000/SSB position and velocity at the same TDB epoch. M2 uses the Sun's GM supplied by the loaded kernel as its authoritative $\mu_\odot$.

2.3 Gravitational acceleration: inertial and Sun-relative equations

The physically transparent inertial, barycentric *direct-force* form for a test satellite is

$$\ddot{\mathbf{x}}_s = \sum_{j \in \mathscr{B}} \mu_j \frac{\mathbf{x}_j(t) - \mathbf{x}_s(t)}{\left\|\mathbf{x}_j(t) - \mathbf{x}_s(t)\right\|^3} + \mathbf{a}_{\mathrm{SRP}}, \tag{2.5}$$

Definitions. $\mathscr{B}$ is the selected set of massive gravitating bodies (including the Sun); $\mu_j = GM_j$ is body j's gravitational parameter [$\mathrm{m^3\ s^{-2}}$]; $\mathbf{a}_{\mathrm{SRP}}$ is radiation-pressure acceleration [$\mathrm{m\ s^{-2}}$].

This is a legitimate equation in an inertial frame *if the motion of every selected massive body is dynamically consistent with the same force set.* Prescribing the Sun's full DE440 trajectory while applying only a subset of the planetary direct accelerations is not equivalent to a consistent subset model: an unpaired part of the Sun's acceleration remains hidden in the subtraction.

2.3.1 Derivation of the implemented heliocentric indirect term

Define satellite and perturber displacement relative to the Sun:

$$\mathbf{r} = \mathbf{x}_s - \mathbf{x}_\odot, \qquad \mathbf{R}_j = \mathbf{x}_j - \mathbf{x}_\odot. \tag{2.6}$$

Differentiate twice:

$$\ddot{\mathbf{r}} = \ddot{\mathbf{x}}_s - \ddot{\mathbf{x}}_\odot. \tag{2.7}$$

For any selected perturber $j \neq \odot$, the satellite's direct acceleration is

$$\mathbf{a}_{j,s} = \mu_j \frac{\mathbf{R}_j - \mathbf{r}}{\left\|\mathbf{R}_j - \mathbf{r}\right\|^3}, \tag{2.8}$$

whereas the same body's acceleration of the Sun is

$$\mathbf{a}_{j,\odot} = \mu_j \frac{\mathbf{R}_j}{R_j^3}, \qquad R_j = \left\|\mathbf{R}_j\right\|. \tag{2.9}$$

Subtracting gives the Sun-relative tidal contribution

$$\mathbf{a}_{j,\mathrm{rel}} = \mu_j \left[\frac{\mathbf{R}_j - \mathbf{r}}{\left\|\mathbf{R}_j - \mathbf{r}\right\|^3} - \frac{\mathbf{R}_j}{R_j^3} \right]. \tag{2.10}$$

The first bracket term is the *direct* planetary attraction on the spacecraft. The second is the *indirect term*, required because a coordinate system whose origin follows the Sun is not strictly inertial. Ignoring it can create fictitious orbital drift.

The integrated M2 equation is therefore

$$\boxed{\ddot{\mathbf{r}} = -\frac{\mu_\odot - k_{\mathrm{SRP}}}{r^3}\mathbf{r} + \sum_{j \neq \odot} \mu_j \left[\frac{\mathbf{R}_j - \mathbf{r}}{\left\|\mathbf{R}_j - \mathbf{r}\right\|^3} - \frac{\mathbf{R}_j}{R_j^3} \right].} \tag{2.11}$$

Definitions. $\mathbf{r}$ is the heliocentric satellite position [m], $\mathbf{R}_j$ heliocentric position of body j [m], $r = \|\mathbf{r}\|$, $R_j = \left\|\mathbf{R}_j\right\|$, $\mu_\odot$ and μ_j gravitational parameters, and k_{SRP} an effective solar-GM reduction [m^3 s^{-2}] derived below. The sum is over the selected planetary perturbers only.

The implemented default list contains the eight planetary systems, treating Earth and Moon separately rather than lumping the Earth–Moon system into one point. Each direct term and its indirect partner use the *same* body subset. Output trajectories are shifted to SSB with Eq. (2.4).

2.4 Deriving solar-radiation-pressure acceleration

A photon carries momentum $p_\gamma = E_\gamma/c$. If an area intercepts electromagnetic energy flux S [W m^{-2}], the normal-incidence momentum flux is approximately S/c [N m^{-2}] for absorption. Reflection changes the momentum transfer; use a configurable coefficient C_r.

$$P_{\mathrm{rad}}(r) = \frac{S_{1\mathrm{au}}}{c}\left(\frac{\mathrm{au}}{r}\right)^2, \tag{2.12}$$

$$\mathbf{a}_{\mathrm{SRP}} = P_{\mathrm{rad}}(r) C_r \frac{A_{\mathrm{eff}}}{m_s}\hat{\mathbf{r}}, \qquad \hat{\mathbf{r}} = \frac{\mathbf{r}}{r}. \tag{2.13}$$

Definitions. P_{rad} is sunlight radiation pressure [N m^{-2}]; $S_{1\mathrm{au}}$ is solar irradiance at 1 au; A_{eff} is effective intercepting/reflecting area [m^2]; m_s is spacecraft mass [kg]; C_r is a dimensionless reflectivity coefficient (approximately 1 for normal-incidence absorption, 2 for ideal specular reflection); $\hat{\mathbf{r}}$ points away from the Sun.

The A_{eff}/m_s ratio is crucial for large, lightweight collecting structures: acceleration rises linearly with area and inversely with mass. Pressure itself decreases as $1/r^2$, just like solar gravity.

2.4.1 Exactly equivalent reduced-gravity parameter

Using Eq. (2.12), collect every position-independent coefficient into

$$k_{\mathrm{SRP}} = \frac{S_{1\mathrm{au}}\mathrm{au}^2}{c} C_r \frac{A_{\mathrm{eff}}}{m_s}, \quad \mathbf{a}_{\mathrm{SRP}} = \frac{k_{\mathrm{SRP}}}{r^3}\mathbf{r}, \quad \mu_{\mathrm{eff}} = \mu_\odot - k_{\mathrm{SRP}}. \tag{2.14}$$

Definitions. k_{SRP} has the same units as $\mu_\odot$, $\mathrm{m}^3\ \mathrm{s}^{-2}$; μ_{eff} is an effective central-force parameter used *only* for this radial SRP approximation.

The solar-gravity plus radial-SRP term is exactly $-\mu_{\mathrm{eff}}\mathbf{r}/r^3$. This identity enables a powerful test: a Sun-only L1 numerical propagation with radial SRP must agree with an M1 analytic Kepler solution using μ_{eff}.

Define the *lightness parameter*

$$\beta_{\mathrm{SRP}} = \frac{k_{\mathrm{SRP}}}{\mu_\odot}. \tag{2.15}$$

Definitions. β_{SRP} is dimensionless and measures outward radiation acceleration relative to inward solar gravity at any radius in this idealization.

For the baseline $C_r = 1.3$ and $A_{\mathrm{eff}}/m_s = 0.02\ \mathrm{m}^2\ \mathrm{kg}^{-1}$, the source documentation reports $\beta_{\mathrm{SRP}} \approx 1.99\times10^{-5}$. Small is not synonymous with irrelevant: secular phase drift can accumulate over a year.

2.4.2 Secular along-track drift estimate

For a nearly circular orbit initialized at the gravitational circular speed, replacing $\mu_\odot$ by $\mu_{\mathrm{eff}} = \mu_\odot(1-\beta_{\mathrm{SRP}})$ slightly changes the effective orbit and mean motion. A first-order estimate is

$$\Delta s \simeq 2\beta_{\mathrm{SRP}}\, n\, t\, a, \qquad \beta_{\mathrm{SRP}} \ll 1,\ e \ll 1,\ 2\beta_{\mathrm{SRP}} nt \ll 1. \tag{2.16}$$

Definitions. Δs is the approximate accumulated along-track separation [m] from an SRP-free orbit; n is M1 mean motion [rad s^{-1}]; t is elapsed duration [s]; a is the approximate orbital radius [m]; e is eccentricity.

The source reports $\Delta s \sim 3.74 \times 10^7$ m after a year for the baseline parameter combination, showing why omitting SRP for low-mass collectors could alter beam targeting substantially.

Important SRP approximation

The implemented SRP force is always radial. A physical flat plate tilted relative to the Sun would have reduced intercepted flux and a possibly nonradial reflection force. A simple absorbing/specular model with incidence angle θ and reflectance ρ gives $\mathbf{F} = P_{\mathrm{rad}} A \cos\theta[(1-\rho)\hat{\mathbf{r}} + 2\rho\cos\theta\hat{\mathbf{n}}]$, with $\hat{\mathbf{n}}$ the illuminated surface normal. This orientation-dependent law is *not* used in the L1 integration; it belongs to later L3 sensitivity work. The implemented C_r matches the normal-incidence coefficient of that idealization.

2.5 Numerical first-order system and DOP853

An ODE integrator evolves a first-order state rather than a second-order equation. With

$$\mathbf{y}(t) = \begin{bmatrix}\mathbf{r}(t)\\ \mathbf{v}(t)\end{bmatrix}, \qquad \frac{\mathrm{d}\mathbf{y}}{\mathrm{d}t} = \begin{bmatrix}\mathbf{v}\\ \mathbf{a}_{\mathrm{rel}}(\mathbf{r},t)\end{bmatrix}, \tag{2.17}$$

Definitions. $\mathbf{y}$ is the six-component state; $\mathbf{a}_{\mathrm{rel}}$ is the right-hand side of Eq. (2.11).

The implementation uses SciPy's adaptive DOP853 solver, an eighth-order explicit Runge–Kutta method with built-in error estimates and dense output. The configured tolerances control local error componentwise, schematically

$$|\delta y_i| \lesssim \mathrm{atol}_i + \mathrm{rtol}\,|y_i|. \tag{2.18}$$

Definitions. δy_i is the estimated local numerical error in state component i; rtol is dimensionless relative tolerance; atol_i is the matching absolute tolerance in metres for a position component or m/s for a velocity component.

Baseline values: $\mathrm{rtol} = 10^{-12}$; $\mathrm{atol}_{\mathbf{r}} = 10^{-3}$ m; $\mathrm{atol}_{\mathbf{v}} = 10^{-9}$ m/s. The configured `integrator_step_s` is a *maximum accepted integration step*, not the fixed output sampling cadence. For the example run it is 300 s; the ephemeris interpolation cache defaults to a separate 3600 s interval.

2.6 Piecewise cubic Hermite ephemeris cache

Repeated SPICE calls inside each ODE evaluation would be expensive. The repository interpolates each body's state between cached $(\mathbf{x}, \mathbf{v})$ endpoints. If $t \in [t_0, t_1]$, let $h = t_1 - t_0$ and $s = (t - t_0)/h$. The cubic Hermite interpolant is

$$\tilde{\mathbf{x}}(t) = H_{00}(s)\mathbf{x}_0 + hH_{10}(s)\mathbf{v}_0 + H_{01}(s)\mathbf{x}_1 + hH_{11}(s)\mathbf{v}_1, \tag{2.19}$$

$$H_{00} = 2s^3 - 3s^2 + 1, \qquad H_{10} = s^3 - 2s^2 + s, \tag{2.20}$$

$$H_{01} = -2s^3 + 3s^2, \qquad H_{11} = s^3 - s^2. \tag{2.21}$$

Definitions. $\tilde{\mathbf{x}}$ is interpolated position [m]; $\mathbf{x}_0, \mathbf{x}_1$ are exact provider endpoint positions [m]; $\mathbf{v}_0, \mathbf{v}_1$ endpoint velocities [m/s]; h is interval duration [s]; s is normalized time [dimensionless]; H_{ij} are dimensionless Hermite basis polynomials.

The interpolation matches both endpoint position and endpoint velocity. Its position error is fourth order in interval width:

$$\|\mathbf{x}(t) - \tilde{\mathbf{x}}(t)\| \lesssim \frac{h^4}{384} \max_{t\in[t_0,t_1]} \left\|\mathbf{x}^{(4)}(t)\right\|. \tag{2.22}$$

Definitions. $\mathbf{x}^{(4)}$ is the fourth derivative of position (m s^{-4}), assumed bounded on the interval. This formula is a conventional componentwise Hermite error scale, not a universal vector-norm guarantee independent of component correlations.

The source measures midpoint differences against direct SPICE. It reports roughly 0.060 m maximum Mercury-barycenter cache error in the tested 2030 year for a 3600 s cache step. A 7200 s cap was adopted because larger intervals make inner-body interpolation unreliable at the targeted accuracy.

2.7 Independent validation and omitted forces

Verification / interpretation

Check 1: With planets and SRP disabled, M2 must reproduce M1 Kepler trajectories and conservation of two-body energy. **Check 2**: With radial SRP enabled but planets off, compare with M1 using μ_{eff}. **Check 3**: Tighten integrator tolerance and step; trajectories must converge. **Check 4**: Compare planetary states with JPL Horizons, and M2 satellite trajectories with REBOUND IAS15. The repository reports a full-model REBOUND difference of about 1.03 m over one year (for its reference run), plus a separately quantified ephemeris-cache error.

The numerical trajectory is precise *within its force model*, but the force model is incomplete. The source documents an omitted first-post-Newtonian solar correction, schematically

$$\mathbf{a}_{1\mathrm{PN}} = \frac{\mu_\odot}{c^2 r^3}\left[\left(\frac{4\mu_\odot}{r} - v^2\right)\mathbf{r} + 4(\mathbf{r}\cdot\mathbf{v})\mathbf{v}\right], \tag{2.23}$$

Definitions. $\mathbf{a}_{1\mathrm{PN}}$ is the weak-field relativistic correction [m/s^2] in the stated coordinate model; r, $\mathbf{r}$, v, $\mathbf{v}$ refer to the Sun-relative state. It is *not* included in the implemented L1 propagator.

For a near-circular 1 AU orbit, its omission produces a reported roughly 56 km/year trajectory difference under the source's comparison conditions. Pluto, selected asteroids, planetary oblateness, detailed attitude-dependent radiation force, Poynting–Robertson drag and satellite back-reaction are likewise not modeled.

Assumptions and limits

Predictive interpretation. The same ephemeris and force assumptions must be used when interpreting differences between runs. A smaller DOP853 integration error does not remove omitted-force, ephemeris, or future Earth-orientation uncertainty. Large spacecraft with unusual area-to-mass ratios warrant dedicated SRP modeling beyond the radial C_r approximation.

M3 — Earth targeting, visibility, and occultation

Primary implementation source: `ground.py`, `visibility.py`; `docs/milestones/M3.md`; `docs/model_assumptions.md`, M3.

Key idea

M3 converts a satellite trajectory into a physically meaningful Earth link. The receiver is attached to a rotating, orbiting Earth; the beam must hit its *future* position after finite light travel time, and the ray must clear Earth, Sun, Moon, and user-defined sky constraints.

3.1 Ground-site geodetic coordinates on WGS 84

A receiver is configured by geodetic latitude ϕ, east longitude λ_g, and height h above the reference ellipsoid. The geodetic latitude is the angle between the ellipsoid's outward surface normal and the equatorial plane; it is *not* in general the latitude of the position vector from Earth's center.

For equatorial radius $a_\oplus$ and flattening $f_\oplus$, introduce the first eccentricity squared and prime-vertical radius of curvature:

$$e_\oplus^2 = f_\oplus(2 - f_\oplus), \qquad N(\phi) = \frac{a_\oplus}{\sqrt{1 - e_\oplus^2 \sin^2 \phi}}. \tag{3.1}$$

Definitions. $e_\oplus$ is the WGS-84 first eccentricity [dimensionless]; $f_\oplus$ Earth flattening [dimensionless]; N the prime-vertical curvature radius [m]; and $a_\oplus$ Earth's WGS-84 equatorial radius [m].

The receiver's Earth-fixed Cartesian coordinates are then

$$x_{\text{ef}} = (N + h)\cos\phi\cos\lambda_g, \tag{3.2}$$

$$y_{\text{ef}} = (N + h)\cos\phi\sin\lambda_g, \tag{3.3}$$

$$z_{\text{ef}} = \left[N(1 - e_\oplus^2) + h\right]\sin\phi. \tag{3.4}$$

Definitions. $x_{\text{ef}}, y_{\text{ef}}, z_{\text{ef}}$ are coordinates [m] in the rotating Earth-fixed frame `IAU_EARTH`; ϕ, λ_g are angles [rad]; h is ellipsoidal altitude [m].

3.2 Local up, east, and north unit vectors

In Earth-fixed coordinates, a natural orthonormal local basis is

$$\hat{\mathbf{u}}_{\text{up}} = (\cos\phi\cos\lambda_g, \cos\phi\sin\lambda_g, \sin\phi), \tag{3.5}$$

$$\hat{\mathbf{u}}_{\text{east}} = (-\sin\lambda_g, \cos\lambda_g, 0), \tag{3.6}$$

$$\hat{\mathbf{u}}_{\text{north}} = (-\sin\phi\cos\lambda_g, -\sin\phi\sin\lambda_g, \cos\phi). \tag{3.7}$$

Definitions. The three $\hat{\mathbf{u}}$ are dimensionless unit directions, tied to the receiver's local tangent plane. The

geodetic up vector is normal to the WGS-84 ellipsoid.

The cross-product identity $\hat{\mathbf{u}}_{\text{east}} \times \hat{\mathbf{u}}_{\text{north}} = \hat{\mathbf{u}}_{\text{up}}$ is a useful independent test of handedness and coordinate sign conventions.

3.3 Earth rotation: an Earth-fixed station is not inertially fixed

Let $\mathbf{R}(t)$ be the orthogonal matrix transforming Earth-fixed components into J2000 components. SPICE's `sxform` provides both $\mathbf{R}(t)$ and its time derivative $\dot{\mathbf{R}}(t)$. Then

$$\mathbf{x}_{\text{site}}(t) = \mathbf{x}_{\oplus}(t) + \mathbf{R}(t)\mathbf{x}_{\text{ef}}, \tag{3.8}$$

$$\mathbf{v}_{\text{site}}(t) = \mathbf{v}_{\oplus}(t) + \dot{\mathbf{R}}(t)\mathbf{x}_{\text{ef}}. \tag{3.9}$$

Definitions. $\mathbf{x}_{\text{ef}}$ is the fixed geodetic site vector [m]; $\mathbf{x}_{\oplus}, \mathbf{v}_{\oplus}$ are Earth-center barycentric position and velocity; $\mathbf{R}$ is dimensionless rotation matrix; $\dot{\mathbf{R}}$ has units s^{-1}; and $\mathbf{x}_{\text{site}}, \mathbf{v}_{\text{site}}$ are the receiver's barycentric position and velocity.

For a simple uniformly rotating Earth, the additional receiver velocity is approximately $\boldsymbol{\omega}_{\oplus} \times (\mathbf{R}\mathbf{x}_{\text{ef}})$, with $\boldsymbol{\omega}_{\oplus}$ angular rotation vector [rad/s]. The implemented `IAU_EARTH` rotation uses text-PCK pole and prime-meridian conventions, rather than a complete IERS station/Earth-orientation solution.

The PCK angle relations are, with T_c centuries and d days past J2000,

$$\alpha_0 = 0^\circ - 0.641^\circ T_c, \tag{3.10}$$

$$\delta_0 = 90^\circ - 0.557^\circ T_c, \tag{3.11}$$

$$W = 190.147^\circ + 360.9856235^\circ d. \tag{3.12}$$

Definitions. α_0 is Earth pole right ascension, δ_0 pole declination, and W the prime-meridian rotation angle. The angular units above are *degrees* as written in the PCK definition, and the time arguments are $T_c = t/(36525 \times 86400\,\text{s})$ and $d = t/(86400\,\text{s})$ for TDB time t relative to J2000.

The IAU model's inertial rotation period is approximately 86,164.099 s. The document explains its accumulated rotation-angle discrepancy against an IERS UT1 orientation, about 23 s of terrestrial rotation by the modeled year 2030 under its stated UT1 assumption. This is a *coordinate-model offset*, not an intrinsic inability to use the simulation for geometric research; actual station targeting would require more accurate Earth orientation.

3.4 Finite light-time: target the receiver on arrival

A beam emitted at time t_e cannot arrive instantaneously. Let $t_a = t_e + \tau$ and define the *arrival-time separation*

$$\boldsymbol{\rho}(\tau) = \mathbf{x}_{\text{site}}(t_e + \tau) - \mathbf{x}_s(t_e). \tag{3.13}$$

Definitions. t_e is satellite emission epoch [TDB s], t_a ground arrival epoch [TDB s], $\tau = t_a - t_e$ is one-way flight time [s], and $\boldsymbol{\rho}$ is the emission-to-arrival ray displacement [m].

With straight-line propagation at speed c, the physical consistency condition is

$$\boxed{c\tau = \|\mathbf{x}_{\text{site}}(t_e + \tau) - \mathbf{x}_s(t_e)\|\,.} \tag{3.14}$$

Why is $t_e + \tau$ inside the receiver's position? While the light travels, Earth moves around the Sun and the

station rotates. A beam aimed at the receiver's *emission-time* position will miss its arrival-time position by roughly transverse velocity times the many-minute travel time.

3.4.1 The implemented fixed-point iteration

Start with the instantaneous range $R_0 = \|\mathbf{x}_{\text{site}}(t_e) - \mathbf{x}_s(t_e)\|$ and iterate

$$\tau_0 = \frac{R_0}{c}, \qquad \tau_{k+1} = \frac{\|\mathbf{x}_{\text{site}}(t_e + \tau_k) - \mathbf{x}_s(t_e)\|}{c}. \tag{3.15}$$

Definitions. τ_k is the kth travel-time estimate [s], R_0 instantaneous distance [m], and k iteration index.

The magnitude of the derivative of this update is bounded approximately by $\|\mathbf{v}_{\text{site}}\|/c \ll 1$ for nonrelativistic station motion, so the map is contractive. The code terminates when successive τ values differ by less than its tolerance (default 10^{-9} s) and falls back to a bracketed Brent root solver if needed. The recorded baseline uses about three iterations, yielding a mean light time of roughly 574.5 s.

The geometric photon direction in the barycentric frame is

$$\hat{\mathbf{u}} = \frac{\boldsymbol{\rho}(\tau)}{R}, \qquad R = \left\|\boldsymbol{\rho}(\tau)\right\|. \tag{3.16}$$

Definitions. R is emission-to-arrival path length [m]; $\hat{\mathbf{u}}$ is the dimensionless unit direction from transmitter at emission to receiver at arrival.

3.5 Relativistic aberration of transmitter and receiver directions

The geometric barycentric ray is not necessarily the emission direction in the moving satellite's rest frame. For relative velocity $\mathbf{v}_s$, let

$$\boldsymbol{\beta} = \frac{\mathbf{v}_s}{c}, \quad \beta = \left\|\boldsymbol{\beta}\right\|, \quad \gamma = \frac{1}{\sqrt{1-\beta^2}}, \quad \hat{\mathbf{b}} = \frac{\boldsymbol{\beta}}{\beta}. \tag{3.17}$$

Definitions. $\boldsymbol{\beta}$ is dimensionless satellite velocity divided by light speed; β its magnitude; γ the Lorentz factor; $\hat{\mathbf{b}}$ the velocity direction (use its continuous limiting formula for $\beta = 0$). These are not M2's β_{SRP}.

The implemented special-relativistic aberration map is

$$\hat{\mathbf{u}}' = \frac{\hat{\mathbf{u}} + (\gamma - 1)(\hat{\mathbf{u}} \cdot \hat{\mathbf{b}})\hat{\mathbf{b}} - \gamma\boldsymbol{\beta}}{\gamma(1 - \boldsymbol{\beta} \cdot \hat{\mathbf{u}})}. \tag{3.18}$$

Definitions. $\hat{\mathbf{u}}'$ is the photon direction in the transmitter's instantaneous rest frame, with axes parallel to J2000.

For $\beta \ll 1$ this becomes

$$\hat{\mathbf{u}}' \simeq \hat{\mathbf{u}} - \boldsymbol{\beta} + (\hat{\mathbf{u}} \cdot \boldsymbol{\beta})\hat{\mathbf{u}}. \tag{3.19}$$

The correction is the *transverse component* of velocity divided by c. Its scale is $v/c \sim 10^{-4}$ rad for heliocentric orbital speeds, vastly larger than the angular width of a narrow optical beam. A separate aberration transform using $\mathbf{v}_{\text{site}}$ provides the apparent incoming direction in the receiver's rest frame.

3.6 Pointing-rate derivative and range rate

Let $\tau'(t_e) = \mathrm{d}\tau/\mathrm{d}t_e$. Differentiate $c\tau = R = \|\boldsymbol{\rho}\|$, where the arrival time depends on emission time:

$$\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t_e} = \mathbf{v}_{\mathrm{site}}(t_a)(1+\tau') - \mathbf{v}_s(t_e), \tag{3.20}$$

$$\tau' = \frac{\hat{\mathbf{u}}\cdot(\mathbf{v}_{\mathrm{site}} - \mathbf{v}_s)}{c - \hat{\mathbf{u}}\cdot\mathbf{v}_{\mathrm{site}}}. \tag{3.21}$$

Definitions. τ' is dimensionless time-dilation of flight time with emission time; $\mathbf{v}_{\mathrm{site}}$ is evaluated at arrival t_a and $\mathbf{v}_s$ at emission t_e.

Differentiating the unit direction yields the exact straight-ray pointing-rate identity

$$\frac{\mathrm{d}\hat{\mathbf{u}}}{\mathrm{d}t_e} = \frac{(\mathbf{I} - \hat{\mathbf{u}}\hat{\mathbf{u}}^{\mathsf{T}})\dot{\boldsymbol{\rho}}}{R}, \qquad \dot{\theta}_{\mathrm{point}} = \|\mathrm{d}\hat{\mathbf{u}}/\mathrm{d}t_e\|\,. \tag{3.22}$$

Definitions. $\mathbf{I}$ is the 3×3 identity matrix; $\hat{\mathbf{u}}\hat{\mathbf{u}}^{\mathsf{T}}$ projects onto the ray direction; $\dot{\theta}_{\mathrm{point}}$ is angular pointing speed [rad/s]. The matrix difference removes longitudinal range changes, leaving only angular motion.

The relative radial velocity of site and spacecraft is

$$\dot{R}_{\mathrm{rel}} = \hat{\mathbf{u}}\cdot(\mathbf{v}_{\mathrm{site}}(t_a) - \mathbf{v}_s(t_e)), \tag{3.23}$$

Definitions. $\dot{R}_{\mathrm{rel}}$ is source-to-receiver range rate [m/s], positive for separation; it is not identically $\mathrm{d}R/\mathrm{d}t_e$ because arrival time depends on emission time.

3.7 Elevation and azimuth at the receiver

Let $\hat{\mathbf{s}}$ be the apparent unit vector from the receiver toward the arriving satellite beam, obtained by *reversing* and aberrating the photon direction in the receiver frame. Rotate the local up/east/north vectors into that same frame. Then

$$\mathrm{el} = \mathrm{atan2}\left(\hat{\mathbf{s}}\cdot\hat{\mathbf{u}}_{\mathrm{up}}, \left\|\hat{\mathbf{s}} - (\hat{\mathbf{s}}\cdot\hat{\mathbf{u}}_{\mathrm{up}})\hat{\mathbf{u}}_{\mathrm{up}}\right\|\right), \tag{3.24}$$

$$\mathrm{az} = \mathrm{atan2}(\hat{\mathbf{s}}\cdot\hat{\mathbf{u}}_{\mathrm{east}}, \hat{\mathbf{s}}\cdot\hat{\mathbf{u}}_{\mathrm{north}}). \tag{3.25}$$

Definitions. el is elevation above local horizon [rad], az azimuth measured from north toward east [rad], and $\hat{\mathbf{s}}$ the incoming apparent unit sightline at the receiver.

The link fails minimum elevation if $\mathrm{el} < \mathrm{el}_{\mathrm{min}}$. It fails Sun avoidance if the apparent satellite–Sun angular separation is less than a configured limit ψ_{min}. The source's optical MVP uses $\mathrm{el}_{\mathrm{min}} = 10^\circ$ and $\psi_{\mathrm{min}} = 0$, the latter meaning *no Sun-avoidance constraint.*

3.8 Beam occultation as a finite-segment geometry problem

The photon travels along a straight segment between transmitter at t_e and receiver at t_a. A moving blocking body B is compared in its own comoving frame. With

$$\mathbf{p}_0 = \mathbf{x}_s(t_e) - \mathbf{x}_B(t_e), \quad \mathbf{p}_1 = \mathbf{x}_{\mathrm{site}}(t_a) - \mathbf{x}_B(t_a), \tag{3.26}$$

Definitions. $\mathbf{p}_0$ and $\mathbf{p}_1$ are ray-endpoint coordinates relative to the moving blocker [m]; B is Earth, Sun or

Moon.

interpolate $\mathbf{p}(q) = \mathbf{p}_0 + q(\mathbf{p}_1 - \mathbf{p}_0)$ for $0 \leqslant q \leqslant 1$. A spherical blocker of radius R_B occludes the ray when

$$\min_{q\in[0,1]} \left\|\mathbf{p}(q)\right\| < R_B. \tag{3.27}$$

Definitions. q is dimensionless ray-segment fraction, not orbital perihelion; R_B is the relevant body radius [m]. A hit must lie on the *segment*, not merely the infinite line.

3.8.1 Earth: ellipsoid rather than sphere

Earth is represented by the WGS-84 ellipsoid

$$\frac{x^2 + y^2}{a_\oplus^2} + \frac{z^2}{b_\oplus^2} = 1, \qquad b_\oplus = a_\oplus(1 - f_\oplus), \tag{3.28}$$

Definitions. $b_\oplus$ is polar semiminor axis [m]; (x, y, z) are Earth-fixed Cartesian coordinates.

Transform the ray endpoints to the Earth-fixed frame and normalize each coordinate by the matching axis. A ray intersects the ellipsoid exactly when the transformed segment enters the unit sphere. For a surface receiver, the final one metre of the segment is excluded in the baseline occultation test to avoid classifying the receiver's own surface endpoint as an obstruction. The code also handles receivers at negative ellipsoidal altitude using an ellipsoid scaled to pass through the receiver.

3.8.2 Sun and Moon

The Sun is modeled as an opaque nominal sphere, with an optional angular solar-limb margin $m_\odot$. An effective shadow radius at Earth–Sun separation $d_\odot$ is

$$R_{\text{block}} = d_\odot \sin\left[\min\left(\arcsin\frac{R_\odot}{d_\odot} + m_\odot, \frac{\pi}{2}\right)\right]. \tag{3.29}$$

Definitions. R_{block} is the effective minimum ray distance from Sun center that constitutes blockage [m]; $d_\odot$ is receiver–Sun distance [m]; $m_\odot$ is configured angular margin [rad].

The Moon is tested with nominal radius R_{Moon}; its position is evaluated consistently during the light flight rather than held fixed at one epoch. A foreground spacecraft nearly aligned with the Sun is not necessarily *occulted* by the solar disk, but may fail the separate receiver Sun-avoidance constraint.

3.9 Solar eclipse and penumbra at the spacecraft

Unlike the previous geometric blockage test for the transmitted beam, spacecraft illumination asks how much of the Sun's apparent disk is obscured by Earth or Moon. Let α and β_B be apparent angular radii of the Sun and a closer occulter, and let d_θ be center-to-center angular separation. Under the small-angle uniform-disk approximation, the overlapped disk area A_{ov} is

$$A_{\text{ov}} = \alpha^2 \cos^{-1}\frac{d_\theta^2 + \alpha^2 - \beta_B^2}{2d_\theta\alpha} + \beta_B^2 \cos^{-1}\frac{d_\theta^2 + \beta_B^2 - \alpha^2}{2d_\theta\beta_B} \tag{3.30}$$

$$- \frac{1}{2}\sqrt{(-d_\theta + \alpha + \beta_B)(d_\theta + \alpha - \beta_B)(d_\theta - \alpha + \beta_B)(d_\theta + \alpha + \beta_B)} \tag{3.31}$$

when the disks partially overlap; set $A_{\mathrm{ov}} = 0$ when separated and equal to the smaller complete disk area when one lies inside the other. Define illumination fraction

$$f_{\mathrm{illum}} = 1 - \frac{A_{\mathrm{ov}}}{\pi\alpha^2}, \qquad 0 \leqslant f_{\mathrm{illum}} \leqslant 1. \tag{3.32}$$

Definitions. A_{ov} [rad^2] is overlap area in the tangent-plane angular approximation, α [rad] the Sun's apparent angular radius, β_B [rad] the blocker radius, d_θ [rad] center separation, and f_{illum} the visible fraction of the Sun's disk.

A full eclipse gives $f_{\mathrm{illum}} = 0$, no eclipse gives 1, and a penumbra gives an intermediate value. M4 multiplies solar flux by this factor. The implementation approximates both disks as uniformly bright and sums separate Earth/Moon obscurations when necessary; overlapped blockers make that sum an upper bound on lost illumination.

3.10 Access fractions, aliasing, and interpretation

Define the binary geometric access indicator

$$g_k = \mathbf{1}\{\mathrm{el}_k \geqslant \mathrm{el}_{\mathrm{min}} \ \wedge \ \text{no selected occultation} \ \wedge \ \text{Sun avoidance passed}\}. \tag{3.33}$$

Definitions. g_k is 1 for a geometrically accessible epoch k and 0 otherwise; $\mathbf{1}\{\cdot\}$ is the logical indicator; $\wedge$ means logical AND.

An access fraction is a weighted or sampled mean $\sum_k w_k g_k / \sum_k w_k$, with weights w_k defined fully in M6. An epoch with access can still have *zero grid power* if M4 thermal constraints or M5/M6 operating conditions fail.

A crucial numerical issue is *diurnal aliasing*: if the sampling step divides approximately a synodic day, every day may be sampled at nearly the same hour angles. The apparent fraction can therefore be biased even though the orbit integrator itself is accurate. The recorded one-hour baseline gives approximately 0.422 access, while an incommensurate 3896 s cadence yields about 0.442.

Verification / interpretation

M3 validation. Geodetic-to-Cartesian conversion is checked against independent WGS-84 implementations; a receiver completes one modeled sidereal turn; synthetic stationary/moving geometries reproduce analytic flight time; Earth intersections match SPICE; Sun/Moon limb cases toggle at the expected boundaries; and pointing-rate derivatives agree with finite differences. The output carries the range, elevation, occultation, light-time, pointing, and orbit-drift diagnostics.

Assumptions and limits

Known physical omissions. The ray is straight in a barycentric frame, with gravitational light bending and Shapiro time delay omitted; the M3 source estimates the latter at about 1.3×10^{-5} s for the baseline geometry. Earth's orientation is an IAU text-PCK approximation; atmosphere and terrain do not refract the geometric ray; illumination uses uniform disks and a small-angle overlap formula; and solar limb and occultation margins are configurable idealizations.

M4 — Solar conversion and thermal equilibrium

Primary implementation source: `solar.py`, `power_chain.py`, `thermal.py`; `docs/milestones/M4.md`; `docs/model_assumptions.md`, M4.

Key idea

M4 answers two coupled questions: How much electrical power can the collector generate at a given Sun distance, and can its surfaces reject the unused energy? The implementation models a photovoltaic (PV) plate and a separate electronics/transmitter radiator as two steady-state radiative nodes with one consistent energy ledger.

4.1 Solar irradiance is an inverse-square flux

If the Sun emits luminosity $L_\odot$ uniformly into space, the total solar energy crossing a sphere of radius $r_\odot$ per unit time is $L_\odot$. The sphere has area $4\pi r_\odot^2$, so energy conservation gives

$$S(r_\odot) = \frac{L_\odot}{4\pi r_\odot^2} = S_{1\mathrm{au}} \left(\frac{\mathrm{au}}{r_\odot} \right)^2 . \tag{4.1}$$

Definitions. $L_\odot$ is solar bolometric luminosity [W] (an explanatory intermediate, not explicitly needed by the implementation); S is irradiance [W m^{-2}]; $r_\odot$ is spacecraft–Sun center distance [m]; $S_{1\mathrm{au}}$ is the configurable solar irradiance at 1 au, with default 1361 W m^{-2}.

The inverse-square law is exact for isotropic outward radiative transport outside a spherical source in the idealized model. The code neglects solar-cycle variability, spectral effects, relative-motion Doppler corrections and collector scattering.

4.2 Projected collection area and partial illumination

A planar photovoltaic array collects the normal component of incident flux. Let $\hat{\mathbf{n}}$ be its illuminated-face unit normal and $\hat{\mathbf{s}}$ the unit direction from the satellite toward the Sun. Then the incidence factor is

$$C_{\mathrm{inc}} = \max(\hat{\mathbf{n}} \cdot \hat{\mathbf{s}}, 0), \tag{4.2}$$

Definitions. $C_{\mathrm{inc}} = \cos\theta$ for a front-facing panel; θ is angle between the panel front normal and the direction to the Sun. Negative values are replaced by zero because the back face has no cells in the model.

The solar power intercepted by the panel is

$$P_{\mathrm{inc}} = S(r_\odot) A_c C_{\mathrm{inc}} f_{\mathrm{illum}}. \tag{4.3}$$

Definitions. P_{inc} is intercepted sunlight power [W]; A_c is collector geometric area [m^2]; f_{illum} is M3's visible-Sun fraction from 0 to 1; $S(r_\odot)$ is flux normal to the sunlight direction.

In `sun_tracking` mode the normal continually follows the Sun, so $C_{\text{inc}} = 1$ in all epochs. This is explicitly an optimistic bound on collectable power. In `fixed_normal` mode the J2000 normal is constant and C_{inc} changes with geometry.

4.3 Absorption, reflection, and what becomes heat

The model treats the opaque collector as absorbing fraction α and rejecting the rest. It therefore partitions incident energy into

$$P_{\text{abs}} = \alpha P_{\text{inc}}, \qquad P_{\text{refl}} = (1-\alpha)P_{\text{inc}}, \qquad P_{\text{abs}} + P_{\text{refl}} = P_{\text{inc}}. \tag{4.4}$$

Definitions. $\alpha \in [0,1]$ is solar absorptivity; P_{abs} and P_{refl} are absorbed and rejected/reflected powers [W]. The $1-\alpha$ term represents all nonabsorbed light in this opaque-collector model.

Reflected photons leave the spacecraft and should *not* be counted as internal waste heat. Conversely, absorbed solar energy not exported electrically must eventually be radiated or stored; the steady-state M4 model assumes no long-term energy storage.

4.4 Temperature-dependent photovoltaic conversion

Define photovoltaic electrical output relative to *absorbed*, not incident, power:

$$P_{\text{PV}}(T_p) = P_{\text{abs}}\,\eta_{\text{PV}}(T_p). \tag{4.5}$$

Definitions. T_p is PV plate temperature [K]; η_{PV} is a dimensionless photovoltaic conversion efficiency; P_{PV} is generated DC electrical power [W].

The repository uses a clipped local linear efficiency law,

$$\eta_{\text{PV}}(T_p) = \text{clip}\Big(\eta_{\text{ref}}[1 + \gamma_T(T_p - T_{\text{ref}})], \eta_{\min}, \eta_{\max}\Big). \tag{4.6}$$

Definitions. η_{ref} is reference efficiency [dimensionless], T_{ref} reference temperature [K], γ_T relative efficiency-temperature coefficient [K^{-1}], and $\eta_{\min}, \eta_{\max}$ admissible efficiency bounds [dimensionless, $0 \leqslant \eta_{\min} < \eta_{\max} \leqslant 1$]. $\text{clip}(x, l, u) = \min(u, \max(l, x))$.

A negative γ_T means hotter PV cells convert a smaller fraction of absorbed light into electricity. The linear law is a *local approximation*, not a general laboratory-validated cell model over 0.3–1.2 AU temperatures. Outside its useful linear regime the function is clamped, and the code flags the clamp.

4.5 The PV thermal node: solve electricity and heat together

The PV plate is approximated as a thin, isothermal panel of physical area A_c, radiating from both faces. Introduce front and back infrared emissivities ϵ_f, ϵ_b, a common view factor F_p toward cold space, and an effective space background temperature T_s. Stefan–Boltzmann exchange then gives

$$Q_{p,\text{rad}}(T_p) = (\epsilon_f + \epsilon_b)\sigma_{\text{SB}} A_c F_p (T_p^4 - T_s^4). \tag{4.7}$$

Definitions. $Q_{p,\text{rad}}$ is radiated PV heat [W]; ϵ_f, ϵ_b are dimensionless longwave emissivities between 0 and 1; F_p is dimensionless view factor; σ_{SB} is the Stefan–Boltzmann constant; T_s is surrounding radiative background temperature [K].

At steady state, absorbed optical energy minus exported electrical energy equals the PV node's own radiation:

$$\boxed{P_{\text{abs}} - P_{\text{abs}}\eta_{\text{PV}}(T_p) = (\epsilon_f + \epsilon_b)\sigma_{\text{SB}}A_cF_p(T_p^4 - T_s^4).} \tag{4.8}$$

This is an *implicit* equation because efficiency depends on the unknown temperature. If the temperature rises, electrical output can fall, which leaves more heat to radiate, feeding back into the temperature. Merely evaluating efficiency at a fixed 300 K would miss this coupling.

4.5.1 Construct the nonlinear residual

Define

$$G_p = (\epsilon_f + \epsilon_b)\sigma_{\text{SB}}A_cF_p, \quad F(T) = P_{\text{abs}}[1 - \eta_{\text{PV}}(T)] - G_p(T^4 - T_s^4). \tag{4.9}$$

Definitions. G_p has units W K^{-4}; $F(T)$ is the PV energy-balance residual [W], equal to zero at equilibrium. The F here is a nonlinear scalar residual, not a radiative view factor.

The scalar root $F(T_p) = 0$ is found using a bracketed Brent solver. This is more robust than unconstrained Newton iteration because the derivative of a clipped efficiency law changes discontinuously at clipping thresholds.

4.5.2 An analytically constructed solver bracket

For an arbitrary fixed efficiency η, the equilibrium temperature has closed form

$$T_{\text{eq}}(\eta) = \left[T_s^4 + \frac{(1-\eta)P_{\text{abs}}}{G_p}\right]^{1/4}. \tag{4.10}$$

Definitions. $T_{\text{eq}}(\eta)$ is the temperature resulting if efficiency were held constant at η.

The code first bounds all possible equilibrium solutions by $[T_s, T_{\text{eq}}(\eta_{\min})]$. Within that interval, it finds the smallest and largest efficiency values, η_{lo} and η_{hi}. Then

$$T_{\text{lo}} = T_{\text{eq}}(\eta_{\text{hi}}), \qquad T_{\text{hi}} = T_{\text{eq}}(\eta_{\text{lo}}) \tag{4.11}$$

brackets the actual root in exact arithmetic. Why? At T_{lo} the true efficiency cannot exceed η_{hi}, making net heat at least the comparison value; at T_{hi} it cannot fall below η_{lo}, making radiative rejection at least the required value. Thus $F(T_{\text{lo}}) \geqslant 0 \geqslant F(T_{\text{hi}})$.

4.5.3 Uniqueness and float64 conditioning

In the unclipped regime,

$$F'(T) = -P_{\text{abs}}\eta_{\text{ref}}\gamma_T - 4G_pT^3. \tag{4.12}$$

For $\gamma_T < 0$, a sufficient uniqueness condition is

$$P_{\text{abs}}\eta_{\text{ref}}|\gamma_T| < 4G_pT_{\text{lo}}^3. \tag{4.13}$$

Definitions. $F'(T)$ is the derivative of the energy residual [W K^{-1}]. The inequality ensures $F'(T) < 0$ throughout the bracket, hence at most one root.

At extremely small absorbed power, directly subtracting $T^4 - T_s^4$ can lose nearly all significant digits. The

code uses the factored identity

$$T^4 - T_s^4 = (T - T_s)(T + T_s)(T^2 + T_s^2) \tag{4.14}$$

and a residual tolerance that includes the unavoidable floating-point representation floor. A small polishing step around the Brent solution handles roots separated by only a few float64 representable temperatures. These are numerical-analysis details, not new physics.

4.6 Electronics and beam-conversion thermal node

Electrical power exported from the PV node passes through three independent on-board conversion stages:

$$P_{\mathrm{dc}} = \eta_{\mathrm{pc}}\, P_{\mathrm{PV}}, \tag{4.15}$$

$$P_{\mathrm{beam}} = \eta_{\mathrm{e2b}}\, P_{\mathrm{dc}}, \tag{4.16}$$

$$P_{\mathrm{ap}} = \eta_{\mathrm{o}}\, P_{\mathrm{beam}}. \tag{4.17}$$

Definitions. P_{dc} is DC available to the transmitter after power conditioning [W]; P_{beam} is electromagnetic beam power before feed/optics loss [W]; P_{ap} leaves the transmitting aperture [W]. Efficiencies η_{pc}, η_{e2b}, and η_{o} represent power conditioning, electrical-to-beam conversion, and optics/RF feed respectively, each in $[0, 1]$.

The *second thermal node* rejects the energy lost in these on-board stages, plus any explicitly configured auxiliary load:

$$\boxed{Q_w = P_{\mathrm{PV}}(1 - \eta_{\mathrm{pc}}) + P_{\mathrm{dc}}(1 - \eta_{\mathrm{e2b}}) + P_{\mathrm{beam}}(1 - \eta_{\mathrm{o}}) + Q_{\mathrm{other}}.} \tag{4.18}$$

Definitions. Q_w is waste heat assigned to the electronics/transmitter radiator [W]; Q_{other} is additional onboard heat load [W].

Why is $P_{\mathrm{abs}} - P_{\mathrm{PV}}$ *not* added to Q_w? Because that is PV plate heat already radiated by Eq. (4.8). Adding it to both would double-count PV heat. The two-node split is a deliberate implementation refinement of the original PDF's more lumped radiator balance.

4.7 Sizing the separate heat radiator

The second thermal node is a thin flat radiator of physical area A_r, emissivity ϵ_r, n_f exposed faces (1 or 2), and free-space view factor F_r. Its outgoing heat is

$$Q_{r,\mathrm{rad}} = \epsilon_r \sigma_{\mathrm{SB}} A_r n_f F_r (T_r^4 - T_s^4). \tag{4.19}$$

Definitions. $Q_{r,\mathrm{rad}}$ is radiator heat rejected to space [W]; A_r is physical radiator surface area [m^2]; ϵ_r infrared emissivity; n_f number of radiating faces; F_r radiative view factor; T_r radiator temperature [K].

With an allowed temperature $T_{r,\max}$, the *minimum area required at that limit* is

$$\boxed{A_{r,\mathrm{req}} = \frac{Q_w}{\epsilon_r \sigma_{\mathrm{SB}} n_f F_r (T_{r,\max}^4 - T_s^4)}.} \tag{4.20}$$

Definitions. $A_{r,\mathrm{req}}$ is the physical radiator area needed [m^2]; $T_{r,\max} > T_s$ is the configured temperature ceiling [K].

If actual area A_r is fixed instead, the equilibrium temperature is

$$T_{r,\mathrm{eq}} = \left[T_s^4 + \frac{Q_w}{\epsilon_r \sigma_{\mathrm{SB}} A_r n_f F_r} \right]^{1/4}. \tag{4.21}$$

A radiator is feasible when $T_{r,\mathrm{eq}} \leqslant T_{r,\max}$. With *automatic sizing*, the model chooses $A_r = A_{r,\mathrm{req}}$ and $T_{r,\mathrm{eq}} = T_{r,\max}$; this is a design requirement, not evidence that the radiator has already been physically integrated into a satellite.

4.8 Feasibility margins and optional radiator mass proxy

Let $T_{p,\max}$ denote the PV maximum operating temperature. Define

$$\Delta T_p = T_{p,\max} - T_p, \tag{4.22}$$

$$\Delta T_r = T_{r,\max} - T_{r,\mathrm{eq}}, \tag{4.23}$$

$$\Delta T_{\mathrm{bind}} = \min(\Delta T_p, \Delta T_r), \tag{4.24}$$

$$\text{feasible} = (\text{PV root converged}) \ \wedge \ (\Delta T_{\mathrm{bind}} \geqslant 0). \tag{4.25}$$

Definitions. ΔT_p is PV thermal margin [K], ΔT_r radiator thermal margin [K], and ΔT_{bind} the binding thermal margin [K]. $\wedge$ denotes logical AND.

If an assumed radiator areal mass density Σ_r [kg m^{-2}] is provided, the radiator mass proxy is simply

$$m_r = A_r \Sigma_r. \tag{4.26}$$

This is a rough proxy: supports, coolant, pointing, shielding, heat pipes and growth margins are not implicitly included.

4.9 The exact energy ledger: track every watt

M4's ledger enforces the following sequence (here $Q_p = Q_{p,\mathrm{rad}}$):

$$P_{\mathrm{inc}} = P_{\mathrm{refl}} + P_{\mathrm{abs}}, \tag{4.27}$$

$$P_{\mathrm{abs}} = P_{\mathrm{PV}} + Q_p, \tag{4.28}$$

$$P_{\mathrm{PV}} = P_{\mathrm{dc}} + L_{\mathrm{pc}}, \tag{4.29}$$

$$P_{\mathrm{dc}} = P_{\mathrm{beam}} + L_{\mathrm{e2b}}, \tag{4.30}$$

$$P_{\mathrm{beam}} = P_{\mathrm{ap}} + L_{\mathrm{o}}, \tag{4.31}$$

$$Q_w = L_{\mathrm{pc}} + L_{\mathrm{e2b}} + L_{\mathrm{o}} + Q_{\mathrm{other}} = Q_{r,\mathrm{rad}}. \tag{4.32}$$

Definitions. $L_{\mathrm{pc}} = P_{\mathrm{PV}} - P_{\mathrm{dc}}$, $L_{\mathrm{e2b}} = P_{\mathrm{dc}} - P_{\mathrm{beam}}$, $L_{\mathrm{o}} = P_{\mathrm{beam}} - P_{\mathrm{ap}}$ are the respective on-board conversion losses [W].

Sum all balances and cancel internal transfers to obtain the spacecraft-boundary identity

$$\boxed{P_{\mathrm{inc}} + Q_{\mathrm{other}} = P_{\mathrm{refl}} + Q_p + Q_{r,\mathrm{rad}} + P_{\mathrm{ap}}.} \tag{4.33}$$

This is the most useful whole-spacecraft energy audit: sunlight and any other supplied heat must leave as reflection, radiation or launched electromagnetic power. Numerically, the ledger closes to a relative

residual of order 10^{-15} in the supplied baseline checks, much tighter than the 10^{-12} acceptance threshold.

4.10 Worked solar/thermal comparison

For the baseline $A_c = 10^4$ m^2, $\alpha = 0.9$, Sun tracking, no eclipse, $\eta_{\mathrm{ref}} = 0.30$, $T_{\mathrm{ref}} = 300$ K, $\gamma_T = -4 \times 10^{-4}$ K^{-1}, and $S_{1\mathrm{au}} = 1361$ W m^{-2}:

$$P_{\mathrm{inc}}(1\mathrm{au}) = 13.61\,\mathrm{MW}, \qquad P_{\mathrm{abs}} = 12.249\,\mathrm{MW}. \tag{4.34}$$

The coupled thermal solution gives about $T_p = 302.8$ K and $P_{\mathrm{PV}} = 3.67$ MW, not the naive exactly 3.675 MW from a fixed 300 K 30% efficiency. With $\eta_{\mathrm{e2b}} = 0.5$ and the remaining onboard efficiencies idealized at unity, the transmitter/electronics radiator rejects about 1.84 MW of heat. The reported required radiator area is about 438.5 m^2 at the specified limit.

At 0.3 AU the characterization still calculates a much larger potential PV output but predicts $T_p \approx 559$ K. Because the assumed PV qualification limit is 400 K, the system policy in M6 refuses to transmit there. This illustrates why *potential electricity* and *operationally deliverable electricity* must be distinguished.

Verification / interpretation

M4 tests. Recover $S(1\mathrm{au})$ and exact inverse-square scaling; force zero PV output for zero illumination or back-facing cells; reproduce analytic radiator scaling $A_r \propto (\epsilon_r T^4)^{-1}$ where $T_s \ll T$; solve coupled PV temperature to verified residual; demonstrate ledger closure; and return a feasibility flag and binding margin for every sample.

Assumptions and limits

Two isothermal steady-state nodes; no thermal inertia or gradients; no conductive or radiative coupling between PV panel and electronics radiator; no direct solar absorption by the radiator; simplified emitter view factors; no temperature-dependent emissivities, PV degradation, irradiance-dependent efficiency, or detailed heat-pipe architecture. The 400 K PV limit is an *assumed scenario bound*, not a material qualification result.

M5 — Diffraction, beam capture, pointing and atmosphere

Primary implementation source: `aperture.py`, `optical.py`, `capture.py`, `microwave.py`, `beam.py`, `atmosphere.py`; `docs/milestones/M5.md`; `docs/model_assumptions.md`, M5.

Key idea

M5 calculates the fraction of transmitted electromagnetic power intercepted by a *finite* terrestrial receiver. It separates (i) the normalized field leaving the aperture, (ii) propagation and diffraction, (iii) receiver projection and pointing, (iv) wavefront and phased-array losses, and (v) atmospheric attenuation. These losses must not be applied twice in M6.

5.1 What the electromagnetic field model assumes

The actual transmitted wave is electromagnetic, but M5 approximates a single polarization component by a complex scalar field. Optical and microwave propagation can be treated with the same mathematical aperture integral once wavelength, aperture illumination and validity regime are specified. The propagator does not model polarization-dependent vector diffraction, plasma and weather-induced wavefront fluctuations, high-power nonlinear optics, or coherent combination of *different* satellites.

5.1.1 Separate the launch power from its spatial distribution

Let D denote the diameter of the transmitting aperture and $a_{\mathrm{ap}} = D/2$ its radius. For a rotationally symmetric launch field $A(r)$, normalize the power *inside the actual aperture*:

$$2\pi \int_0^{a_{\mathrm{ap}}} |A(r)|^2 \, r \, \mathrm{d}r = 1, \qquad A(r) = 0 \quad \text{for } r > a_{\mathrm{ap}}. \tag{5.1}$$

Definitions. r here is *radial distance in the transmitter aperture plane* [m], not heliocentric orbital distance; $A(r)$ is the complex scalar aperture-field amplitude with units m^{-1} after normalization; $\mathrm{d}r$ is the radial differential. A unit-normalized field represents one watt of power through the aperture; the actual outgoing power P_{ap} multiplies the intensity later.

This normalization is the conceptual reason the code does not apply a separate generic "aperture efficiency" during capture: losses before the aperture are represented by η_{o} in M4/M6, while M5 calculates the spatial distribution of whatever power *actually leaves* the aperture.

5.2 Three implemented aperture-illumination profiles

5.2.1 Uniform circular illumination

If the field is constant inside the disk, write $A(r) = C_u$. Normalizing Eq. (5.1) gives

$$|C_u|^2 = \frac{1}{\pi a_{\mathrm{ap}}^2}. \tag{5.2}$$

Definitions. C_u [m^{-1}] is the uniform aperture amplitude. Uniform illumination is the assumption behind the familiar Airy diffraction pattern.

5.2.2 Circularly truncated Gaussian illumination

Let w_0 be the radius at which the *intensity* of a Gaussian is e^{-2} of its on-axis value. Then

$$A_G(r) = C_G e^{-r^2/w_0^2} \quad (r \leqslant a_{\mathrm{ap}}), \tag{5.3}$$

$$|C_G|^{-2} = \frac{\pi w_0^2}{2}\left(1 - e^{-2a_{\mathrm{ap}}^2/w_0^2}\right). \tag{5.4}$$

Definitions. A_G is normalized Gaussian aperture amplitude [m^{-1}]; C_G its normalization constant [m^{-1}]; and w_0 is a Gaussian intensity radius [m].

The fraction of source-Gaussian power outside a finite aperture would be $e^{-2a_{\mathrm{ap}}^2/w_0^2}$. That is *spill-over not launched*, and belongs to the already separated optics/feed efficiency, not an extra multiplier in M5 capture.

5.2.3 Parabolic taper on a nonzero pedestal

For an edge taper of T_{edge} dB in *intensity*, define the relative *field-amplitude* pedestal

$$C = 10^{-T_{\mathrm{edge}}/20}, \qquad A_P(r) = C_P\left[C + (1-C)\left(1 - \frac{r^2}{a_{\mathrm{ap}}^2}\right)\right] \tag{5.5}$$

for $0 \leqslant r \leqslant a_{\mathrm{ap}}$, with normalization

$$|C_P|^{-2} = \pi a_{\mathrm{ap}}^2\left[C + \frac{(1-C)^2}{3}\right]. \tag{5.6}$$

Definitions. T_{edge} is the edge power taper [dB]; C is the edge-to-central field-amplitude ratio [dimensionless]; C_P is a normalization constant [m^{-1}]; and $A_P(r)$ is the tapered field.

Why 20 instead of 10 in the exponent? Decibels here describe a power/intensity ratio, but field amplitude is proportional to the square root of power, so an amplitude ratio uses $10\log_{10}$ on amplitude squared, equivalently $20\log_{10}$ on amplitude.

5.3 Scalar diffraction from a circular aperture

Let λ be free-space wavelength, z propagation distance from aperture to receiver, and ρ radial distance in the receiver plane from the nominal beam axis. A monochromatic scalar field at the receiving plane can be computed using the paraxial Fresnel–Hankel integral:

$$U(\rho; z) = \frac{2\pi}{\lambda z}\exp\left(\frac{i\pi\rho^2}{\lambda z}\right)\int_0^{a_{\mathrm{ap}}} A(r)\exp\left(\frac{i\pi r^2}{\lambda z}\right)J_0\left(\frac{2\pi r\rho}{\lambda z}\right) r\,\mathrm{d}r. \tag{5.7}$$

Definitions. $U(\rho; z)$ is complex receiver-plane amplitude normalized per square-root transmitted watt [m^{-1}]; $i = \sqrt{-1}$ is the imaginary unit (not orbital inclination); J_0 is the order-zero Bessel function of the first kind; ρ [m] is a transverse receive-plane radius; r [m] is transmitter aperture radius coordinate; and z [m] is propagation distance.

The receiver-plane intensity per aperture watt is

$$I(\rho; z) = |U(\rho; z)|^2, \quad I\ [\mathrm{m}^{-2}], \quad 2\pi \int_0^\infty I(\rho; z)\rho\, \mathrm{d}\rho \approx 1. \tag{5.8}$$

The final power-conservation identity follows from the unitarity of paraxial free-space propagation (Hankel Parseval). It is an invaluable numerical test: diffraction may *redistribute* power but must not create it. Multiplying I by outgoing aperture power P_{ap} yields irradiance in $\mathrm{W/m}^2$.

5.3.1 Far-field (Fraunhofer) limit

When propagation distance is large enough, the quadratic phase inside the aperture is negligible for the relevant accuracy. For an observation angle $\theta \simeq \rho/z$, define angular intensity per watt

$$J(\theta) = \frac{1}{\lambda^2}\left|2\pi \int_0^{a_{\mathrm{ap}}} A(r) J_0\left(\frac{2\pi r\theta}{\lambda}\right) r\, \mathrm{d}r\right|^2, \tag{5.9}$$

$$I(\rho; R) = \frac{1}{R^2} J\left(\frac{\rho}{R}\right). \tag{5.10}$$

Definitions. $J(\theta)$ is angular beam power density per emitted watt [sr^{-1}], where sr denotes steradian; R [m] is propagation range; I is irradiance per emitted watt [m^{-2}].

The $1/R^2$ factor follows from geometric expansion of angular solid angle into transverse area. A fixed angular beam becomes physically wider in proportion to R.

5.3.2 The Fresnel number and model-validity decisions

Define

$$N_F = \frac{D^2}{\lambda R}. \tag{5.11}$$

Definitions. N_F is dimensionless Fresnel number; D is transmitter aperture diameter [m]; λ wavelength [m]; R propagation range [m].

The repository permits far-field models only when $N_F \leqslant 0.5$, equivalent to $R \geqslant 2D^2/\lambda$. If this fails, a far-field request raises a validity error; the Fresnel–Hankel solver is used as an alternative where its paraxial approximation is valid.

The finite-distance Fresnel approximation also neglects quartic terms in the optical path. The code estimates the largest omitted phase as

$$\phi_4 \sim \frac{\pi(a_{\mathrm{ap}} + \rho_{\max})^4}{4\lambda z^3} \tag{5.12}$$

and requires it to be below a configured phase threshold, 0.1 rad in the reviewed model. Here $\rho_{\max}$ is the largest transverse receiver/pointing radius at which the field will be used [m]. This tests whether a model's approximation is appropriate *before* trusting its numerical output.

5.4 The Airy pattern: a uniformly illuminated circular aperture

For uniform illumination in the far field, the Bessel integral has a closed-form result. Let

$$x = \frac{\pi D \rho}{\lambda R}, \qquad I_0 = \frac{\pi D^2}{4\lambda^2 R^2}. \tag{5.13}$$

Definitions. x is dimensionless radial diffraction coordinate; I_0 is on-axis irradiance per transmitted watt [m^{-2}].

Then

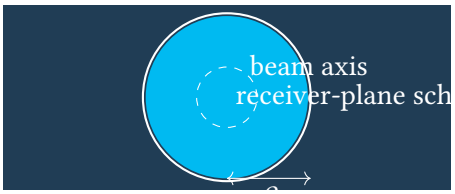
$$\boxed{I_{\mathrm{Airy}}(\rho) = I_0 \left[\frac{2J_1(x)}{x}\right]^2}, \qquad \lim_{x\to 0} \frac{2J_1(x)}{x} = 1. \tag{5.14}$$

Definitions. J_1 is the order-one Bessel function of the first kind; the value at zero is defined by its finite limit.

The first dark ring occurs when $J_1(x)$ first vanishes, at $j_{1,1} \approx 3.831706$. Therefore

$$\rho_{\mathrm{null}} = \frac{j_{1,1}}{\pi}\frac{\lambda R}{D} \approx 1.21967 \frac{\lambda R}{D}. \tag{5.15}$$

Definitions. ρ_{null} is the radial distance from the central beam axis to the first Airy intensity zero [m], and $j_{1,1}$ is the first positive root of J_1.



A finite receiver does not necessarily include the whole bright central spot. Even a receiver large enough to enclose the *first dark ring* captures only about 83.8% of an ideal Airy pattern: appreciable power remains in outer rings.

5.4.1 Encircled energy and the small-receiver limit

For a centered circular disk of radius a_r, define encircled-energy capture

$$\eta_{\mathrm{EE}}(a_r) = 2\pi \int_0^{a_r} I(\rho)\rho \, \mathrm{d}\rho. \tag{5.16}$$

Definitions. a_r [m] is receiver radius projected perpendicular to the beam; η_{EE} is the dimensionless fraction of launched aperture power intercepted when perfectly centered, before atmosphere and receiver conversion.

For the Airy pattern the integral simplifies to

$$\eta_{\mathrm{EE}}(x) = 1 - J_0(x)^2 - J_1(x)^2, \qquad x = \frac{\pi D a_r}{\lambda R}. \tag{5.17}$$

The repository checks $\eta_{\mathrm{EE}}(j_{1,1}) \approx 0.83778$. When $x \ll 1$, series expansion gives

$$\eta_{\mathrm{EE}}(x) \simeq \frac{x^2}{4} = \frac{\pi^2 D^2 a_r^2}{4\lambda^2 R^2}. \tag{5.18}$$

This small-capture scaling is vital: captured fraction grows approximately as D^2, receiver radius squared, and wavelength inverse squared, while falling as range squared.

5.5 Gaussian-beam alternative

For a sufficiently untruncated TEM_{00} Gaussian with waist radius w_0 at the transmitter plane, define Rayleigh range

$$z_R = \frac{\pi w_0^2}{\lambda}, \qquad w(z) = w_0\sqrt{1 + \left(\frac{z}{z_R}\right)^2}. \tag{5.19}$$

Definitions. z_R is Rayleigh range [m]; $w(z)$ is the Gaussian intensity e^{-2} radius at distance z [m]; w_0 is waist radius [m].

The normalized radial intensity is

$$I_G(\rho, z) = \frac{2}{\pi w(z)^2}\exp\left(-\frac{2\rho^2}{w(z)^2}\right), \quad \eta_{G,\mathrm{EE}}(a_r) = 1 - e^{-2a_r^2/w(z)^2}. \tag{5.20}$$

The code permits this *untruncated* Gaussian approximation only for sufficiently weak aperture clipping, baseline $a_{\mathrm{ap}}/w_0 \geqslant 2.5$. For more strongly clipped fields the Fresnel–Hankel integral with an explicitly truncated Gaussian is required. Labeling an arbitrary clipped aperture a Gaussian beam would conceal diffraction sidelobes.

5.6 Off-axis receiver: exact finite-disk interception

A moving or mispointed transmitter places the beam center at transverse offset d from the receiver center. A ring of radius ρ around the beam axis intersects only part of a circular receiver of radius a_r. Define $\varphi(\rho; d, a_r)$ as the half-angle subtended by the ring portion inside the receiver. For partial overlap,

$$\varphi(\rho; d, a_r) = \arccos\left(\mathrm{clip}\left(\frac{\rho^2 + d^2 - a_r^2}{2\rho d}, -1, 1\right)\right). \tag{5.21}$$

Definitions. d is transverse beam-axis displacement from receiver center [m]; φ is half angular span [rad]; ρ is distance from beam axis [m]; a_r receiver radius [m]. Cases $d = 0$ and complete/no overlap use their appropriate limiting formulas.

The intercepted fraction is

$$\boxed{\eta_{\mathrm{disk}}(d) = \int_0^\infty I(\rho)\,[2\varphi(\rho; d, a_r)]\rho\,\mathrm{d}\rho.} \tag{5.22}$$

The factor $2\varphi\rho\mathrm{d}\rho$ is the area of the ring portion contained in the receiving disk. This formula is why calculating merely the beam's center intensity is insufficient: the receiver has finite area and can be offset from the beam center.

The code splits the always-contained inner disk $\rho \leqslant \max(a_r - d, 0)$ from the partial-overlap band $|a_r - d| \leqslant \rho \leqslant a_r + d$, and integrates with Gauss–Legendre quadrature plus an endpoint-smoothing substitution. When $d = 0$, Eq. (5.22) must reduce exactly to encircled energy, Eq. (5.16).

5.7 Receiver orientation and projected geometry

A horizontal circular receiver looks elliptical when viewed along an oblique incoming beam. At elevation el above the local horizon, a disk of physical radius a_r projects onto the beam-normal plane as an ellipse with semiaxes a_r and $a_r \sin \mathrm{el}$. Its area is

$$A_{\mathrm{proj}} = \pi a_r^2 \sin \mathrm{el}. \tag{5.23}$$

To reuse the circular capture integral, M5 approximates that ellipse by an equal-area circle

$$a_{\mathrm{eff}} = a_r \sqrt{\sin \mathrm{el}}. \tag{5.24}$$

Definitions. A_{proj} is projected receiver area [m^2]; a_{eff} is equal-area effective radius [m]. For a receiver mechanically tracking the beam, $a_{\mathrm{eff}} = a_r$.

The equal-area replacement is not exact when the receiver is comparable with the beam spot; the code provides a separate centered-ellipse diagnostic. In particular, a large horizontal receiver at low elevation can be overestimated by the equal-area circular approximation.

5.8 Pointing bias and two-dimensional angular jitter

Let the angular boresight error have fixed magnitude b_θ (bias) plus independent Gaussian random errors of standard deviation σ_θ on each of two orthogonal angular axes. At range R, the transverse error components have standard deviation

$$s = R\sigma_\theta, \qquad \nu_b = Rb_\theta. \tag{5.25}$$

Definitions. σ_θ is per-axis angular pointing jitter [rad]; b_θ is deterministic angular bias [rad]; s is transverse standard deviation [m]; ν_b is fixed transverse bias distance [m].

The magnitude d of the two-dimensional offset follows a *Rice distribution*:

$$f_{\mathrm{Rice}}(d; \nu_b, s) = \frac{d}{s^2} \exp\left(-\frac{d^2 + \nu_b^2}{2s^2}\right) I_0\left(\frac{d\nu_b}{s^2}\right), \qquad d \geqslant 0, \tag{5.26}$$

Definitions. f_{Rice} is probability density with units m^{-1}; I_0 here is the modified Bessel function of the first kind, order zero (distinct from Airy on-axis intensity I_0); ν_b and s are Rice distribution parameters [m].

Expected capture is the probability-weighted finite-receiver integral

$$\overline{\eta}_{\mathrm{diff}} = \mathbb{E}[\eta_{\mathrm{disk}}(d)] = \int_0^\infty \eta_{\mathrm{disk}}(d) f_{\mathrm{Rice}}(d; \nu_b, s)\, \mathrm{d}d. \tag{5.27}$$

Definitions. $\overline{\eta}_{\mathrm{diff}}$ is the expected dimensionless diffraction capture over the pointing-error distribution. Its integrand is capture at a particular offset multiplied by the probability density of that offset.

The implementation uses deterministic numerical quadrature over the significant Rice support, not randomly drawing an offset for each time epoch. It also computes capture percentiles (5th, 50th, 95th) of the modeled offset distribution. These are *conditional pointing-distribution percentiles*, not statistical uncertainty bands on the underlying physical laws.

5.8.1 Pointing efficiency and its caveat

Define

$$\eta_{\text{point}} = \frac{\mathbb{E}[\eta_{\text{disk}}(d)]}{\eta_{\text{disk}}(0)}, \tag{5.28}$$

Definitions. η_{point} is relative performance of the chosen pointing-error distribution against perfectly centered capture, assuming the denominator is nonzero.

For a centered receiver entirely inside the smooth Airy central lobe, raising jitter normally reduces capture. However, ring patterns are nonmonotonic in radius: if a receiver intersects an outer ring, some displacement can shift ring power into the receiver. The source documents counterexamples where $\eta_{\text{point}} > 1$. Therefore it is *not* universally a passive efficiency in the sense of being bounded by one; M6 uses it to enforce an optional pointing-availability threshold, while the actual total capture fraction remains bounded by one.

5.9 Wavefront phase-error reduction

Even with correct mean pointing, an imperfect aperture wavefront reduces coherent concentration into the intended diffraction pattern. If σ_w is RMS *optical path* or wavefront error, define RMS phase error

$$\phi_{\text{rms}} = \frac{2\pi\sigma_w}{\lambda}. \tag{5.29}$$

Definitions. σ_w is RMS wavefront path-length error [m], not mirror surface height; ϕ_{rms} is the corresponding RMS phase error [rad].

The implemented extended Maréchal/Ruze factor is

$$\eta_{\text{wf}} = \exp(-\phi_{\text{rms}}^2) = \exp\left[-\left(\frac{2\pi\sigma_w}{\lambda}\right)^2\right]. \tag{5.30}$$

Definitions. η_{wf} is a dimensionless coherent-core degradation factor, at most one.

The model assumes that the power scattered outside the coherent core is *not recaptured* by the considered receiver. This is an approximation (often conservative for broad halos) rather than full phase-screen wave propagation. It is applied exactly once.

5.10 Microwave phased-array lattice factor

For a square lattice of $n_e \times n_e$ radiating elements spaced by d_e, the one-dimensional array response in an angular direction coordinate u is

$$D_{n_e}(\psi) = \frac{\sin(n_e\psi/2)}{\sin(\psi/2)}, \qquad \psi = \frac{2\pi d_e u}{\lambda}. \tag{5.31}$$

Definitions. n_e is element count along one axis (integer); d_e element separation [m]; u is a dimensionless direction cosine; and ψ is phase increment between adjacent elements [rad].

For two angular direction cosines u, v, the separable array factor is

$$\text{AF}(u, v) = D_{n_e}(2\pi d_e u/\lambda) D_{n_e}(2\pi d_e v/\lambda). \tag{5.32}$$

Grating-lobe directions satisfy $(u, v) = (m, n)\lambda/d_e$ for integers m, n lying in the visible angular domain. These are additional maxima that can send power away from the intended central beam. The repository approximates the fraction of radiated power remaining in the desired main lobe by

$$\eta_{\text{lattice}} = \left(\sum_{(m,n)\in\mathscr{V}} \cos^{q-1} \theta_{mn} \right)^{-1}. \tag{5.33}$$

Definitions. $\mathscr{V}$ is the set of visible grating-lobe orders; θ_{mn} is the angle of order (m, n) from array broadside; and $q \geqslant 1$ is a configured element-pattern exponent. η_{lattice} is dimensionless.

The implementation enforces a sufficiently large array and keeps the evaluated receiver region away from nearby grating lobes. A filled tapered microwave aperture uses the same optical-style Fraunhofer/Fresnel machinery instead.

5.11 Assemble beam capture without loss double-counting

The quantity returned to M6 is

$$\boxed{\eta_{\text{capture}} = \eta_{\text{wf}}\, \eta_{\text{lattice}}\, \overline{\eta}_{\text{diff}}.} \tag{5.34}$$

Definitions. η_{capture} is expected fraction of the power reaching the ground receiver plane that falls on its effective aperture; η_{wf} is wavefront factor; η_{lattice} is microwave lattice main-lobe factor (unity for optical); and $\overline{\eta}_{\text{diff}}$ includes diffraction, receiver projection and pointing statistics.

At no point should η_{capture} also be multiplied by a second taper, pointing, Ruze or aperture-efficiency factor. Atmospheric transmission is *not* part of η_{capture}; it has its own independent stage in M6.

5.12 Atmospheric attenuation on the terminal Earth segment

M5 distinguishes an optical/microwave *scenario* model and a microwave ITU-R empirical model. The atmosphere is treated as a terminal link effect depending on carrier, station and elevation, not on the full interplanetary vacuum range.

5.12.1 Simple atmospheric airmass approximation

For elevation el > 0, plane-parallel atmospheric airmass is

$$X(\text{el}) = \frac{1}{\sin \text{el}}. \tag{5.35}$$

Definitions. X is dimensionless airmass relative to zenith; el is receiver elevation angle [rad].

At zenith, $X = 1$; at lower elevation the beam traverses more atmosphere. This flat-atmosphere approximation diverges at the horizon. The program restricts its simple model to elevations at least 10 degrees by default rather than extrapolating it to the horizon.

For optical zenith transmittance τ_z,

$$\eta_{\text{atm,opt}} = \tau_z^{X(\text{el})}. \tag{5.36}$$

Definitions. $\tau_z \in [0, 1]$ is assumed one-way optical transmittance at zenith; $\eta_{\text{atm,opt}}$ is a dimensionless transmission factor. $\tau_z = 0$ is treated as a hard cloudy outage.

For microwave zenith attenuation A_z in decibels,

$$A(\mathrm{el}) = A_z X(\mathrm{el}), \qquad \eta_{\mathrm{atm,RF}} = 10^{-A(\mathrm{el})/10}. \tag{5.37}$$

Definitions. A_z is assumed zenith power attenuation [dB], $A(\mathrm{el})$ slant attenuation [dB], and $\eta_{\mathrm{atm,RF}}$ the resulting power transmission fraction. The factor 10 comes from the definition of decibels for *power*.

5.12.2 ITU-R microwave model: externally specified attenuation

The `itu_r` option calls the `itur` library for an ITU-R P.618-based statistical slant-path attenuation:

$$A_p = A_{\mathrm{gas,p}} + A_{\mathrm{cloud,p}} + A_{\mathrm{rain,p}}, \qquad \eta_{\mathrm{atm,ITU}} = 10^{-A_p/10}. \tag{5.38}$$

Definitions. A_p [dB] is attenuation exceeded for a configured percentage p of an average year; the three terms are gas, cloud and rain attenuation in dB for the given frequency, elevation, site and polarization assumptions. This is an external standards-based model, *not* an in-repository closed-form derivation of every ITU atmospheric submodel.

The production wrapper pins recommendation versions and excludes scintillation for the kilometre-scale rectenna architecture. Its $p = 1\%$ setting is a long-term statistical design condition applied deterministically; *it is not a simulated random weather time series.* The source documents that certain production revisions lack an independent published numerical reference set in the installed `itur` package, even though regression and older-version example checks pass.

5.13 Physical scaling exercise: why interplanetary capture is difficult

Consider a uniform laser aperture $D = 10$ m at $\lambda = 10^{-6}$ m and range $R \simeq 1.15$au. Equation (5.15) predicts a first dark-ring radius of about 21 km. A circular ground receiver with radius 1 km intercepts only a small part of its central bright disk, explaining the reported median expected capture near 6.3×10^{-3} during available epochs.

The microwave baseline uses $\lambda \approx 0.12236$ m, a filled/tapered 1 km aperture, and a 10 km-radius ground rectenna. The much longer wavelength gives a very broad interplanetary footprint; the numerical median expected capture is near 3.88×10^{-7}. These are different hardware assumptions, not an apples-to-apples technology ranking.

Verification / interpretation

M5 checks. Airy first null equals $1.21967\lambda R/D$; encircled energy at the first dark ring equals 0.83778; the pattern integrates to unit power; finite-disk quadrature agrees with a converged 2-D integration; Gaussian capture agrees with an independent noncentral-χ^2 formula; HCIPy reproduces selected far-field aperture profiles; Fresnel and far-field models refuse invalid regimes; pointing and atmosphere factors stay within the expected physical ranges under the assumptions of each model.

Assumptions and limits

Not modeled. Turbulent beam wander and adaptive optics, laser thermal blooming, detailed receiver optics, wavefront phase-screen halo recapture, finite source linewidth, polarization-dependent vector effects, coherent distributed beamforming, obstruction by transmitter structures, strong-angle diffraction, and a stochastic meteorological process. These omissions can make a calculated link an optimistic upper bound and must be reported in any manuscript.

M6 — End-to-end single-satellite energy delivery

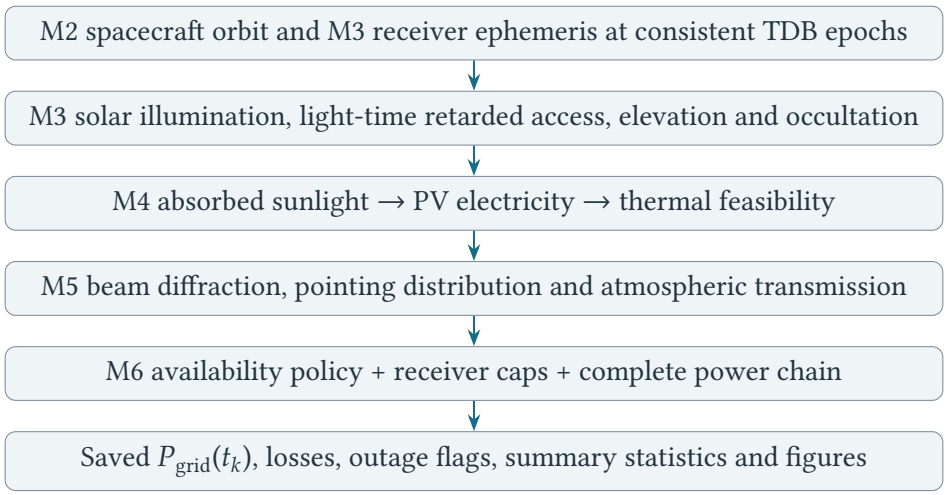
Primary implementation source: `system.py`, `link_budget.py`, `metrics.py`, `io.py`, `manifest.py`, `sweep.py`, `capture_batch.py`; `docs/milestones/M6.md`, M6b audit; `docs/model_assumptions.md`, M6.

Key idea

M6 is the *vertical slice*: it combines M1–M5 into one time-resolved answer to the scientific question, "How many watts arrive at the ground power grid, and how reliably?" The on-board PV output is never confused with received electricity, and all unavailability, receiver caps and losses are applied in a strict order.

6.1 One satellite, one receiver, one time-indexed calculation

The run is specified by a validated and versioned configuration. The baseline has a time grid $\{t_k\}$, a spacecraft state $\{\mathbf{x}_s(t_k), \mathbf{v}_s(t_k)\}$, and one Earth receiver. At each emission epoch, M6 orchestrates the following dependency chain:



All mathematics is evaluated at emission epochs $t_e = t_k$; M3 identifies the arrival time $t_a = t_e + \tau$. Received-energy accounting uses the emission time axis as an approximation, with a documented relative timing/-Doppler correction scale $\sim |\dot{R}|/c \lesssim 10^{-5}$ for the baseline.

6.2 Availability logic must be evaluated before the power budget

For each emission sample k, define a blocking indicator B_k as the logical OR of these ten independent conditions:

1. receiver elevation below the allowed minimum;
2. Earth occultation of the ray;
3. Sun occultation of the ray;

4. Moon occultation of the ray;
5. receiver Sun-avoidance violation;
6. full spacecraft eclipse ($f_{\text{illum}} = 0$);
7. M4 thermal infeasibility;
8. modeled hard weather outage;
9. violation of an optional pointing-efficiency minimum;
10. projected peak irradiance above the configured beam-abort safety limit.

Set

$$a_k = 1 - B_k, \qquad u_k = \mathbf{1}\{\text{commanded transmission AND } a_k = 1\}. \tag{6.1}$$

Definitions. B_k is logical blocking status (0 or 1), a_k is instantaneous availability (0 or 1), and χ_k is the active-transmitting indicator (0 or 1). The precise implementation maintains a bitmask of *all* conditions, not just one failure reason.

The code records the first active blocking bit in a specified priority ordering as a *primary* reason for outage summaries. Separately, it records all overlapping flags. Consequently, the primary fractions add to total unavailability, but overlapping flagged fractions generally do not.

Two additional flags are *nonblocking*: partial eclipse reduces collector power but does not force complete shutdown, while receiver saturation clips delivered power but still permits a link.

Operating policies are modeling choices

M6 turns the transmitter fully off whenever its modeled thermal state is infeasible. It does not search for a lower-power operating point or model the thermal response after switching off. The beam-density constraint is also an *abort*, not power throttling. These policies should be reconsidered during later optimization, but cannot be silently replaced while comparing reported M6 baseline runs.

6.3 The complete algebraic link budget

The program treats each efficiency exactly once. At an epoch t_k, begin with M4 photovoltaic output $P_{\text{PV,k}}$, then evaluate

$$P_{\text{PV,tx}} = \chi_k P_{\text{PV}}, \tag{6.1a}$$
$$P_{\text{dc}} = \eta_{\text{pc}} P_{\text{PV,tx}}, \tag{6.1b}$$
$$P_{\text{beam}} = \eta_{\text{e2b}} P_{\text{dc}}, \tag{6.1c}$$
$$P_{\text{ap}} = \eta_{\text{o}} P_{\text{beam}}, \tag{6.1d}$$
$$P_{\text{atm}} = \eta_{\text{atm}} P_{\text{ap}}, \tag{6.1e}$$
$$P_{\text{capt}} = \eta_{\text{capture}} P_{\text{atm}}, \tag{6.1f}$$
$$P_{\text{accept}} = \min(P_{\text{capt}}, P_{\text{max}}, P_Q), \tag{6.1g}$$
$$P_{\text{receiver}} = \eta_{\text{rx}} P_{\text{accept}}, \tag{6.1h}$$
$$P_{\text{grid}} = \eta_{\text{grid}} P_{\text{receiver}}. \tag{6.1i}$$

Definitions. P_{PV} is M4 PV-generated power even when not transmitting; $P_{\text{PV,tx}}$ is PV power committed to active transmission; P_{dc} is conditioned electrical power; P_{beam} electromagnetic beam power before feed loss; P_{ap} power leaving the transmitter aperture; P_{atm} power remaining after atmospheric attenuation; P_{capt} power intercepted at the receiver; P_{accept} intercepted power within receiver limits; P_{receiver} electricity

converted by laser-PV or rectenna; and P_{grid} electricity delivered after grid conversion. All powers are in watts.

Definitions. $\eta_{\text{pc}}, \eta_{\text{e2b}}, \eta_{\text{o}}$ are M4 on-board efficiencies; η_{atm} is M5 atmospheric transmission; η_{capture} is M5 finite-aperture expected capture including modeled pointing/wavefront/lattice factors; η_{rx} is ground receiver conversion efficiency; η_{grid} is grid-interface efficiency. All are dimensionless fractions in $[0, 1]$.

Definitions. P_{max} is receiver maximum acceptable incident beam power [W]. P_Q is the incident-power limit implied by receiver waste-heat capacity, derived next. The index k is suppressed on subsequent stage powers for readability; each expression applies to a given epoch.

Physical reasoning: each line is either multiplication by a conversion fraction, a binary operating decision, or a receiver saturation cap. Therefore no passive stage may create more energy than its input, and $P_{\text{grid}} = 0$ if $\chi_k = 0$.

6.3.1 Receiver heat-rejection constraint

An optical photovoltaic receiver or rectenna converts only fraction η_{rx} of accepted incident beam power into electricity. The rest becomes receiver waste heat:

$$Q_{\text{rx}} = (1 - \eta_{\text{rx}}) P_{\text{accept}}. \tag{6.2}$$

Definitions. Q_{rx} is ground receiver heat load [W].

For a configured receiver maximum waste-heat capacity Q_{max},

$$P_Q = \begin{cases} Q_{\text{max}}/(1 - \eta_{\text{rx}}), & \eta_{\text{rx}} < 1, \\ +\infty, & \eta_{\text{rx}} = 1. \end{cases} \tag{6.3}$$

Definitions. Q_{max} is allowable ground-receiver heat rejection [W]; P_Q is the corresponding admissible received beam power [W]. An unset receiver limit is likewise treated as unbounded for that constraint.

This limit is applied *before* multiplying by η_{rx}, to avoid incorrectly treating a thermal constraint as an electrical-output cap.

6.3.2 Maximum beam irradiance and beam-abort rule

The beam model supplies a normalized, idealized peak irradiance I_{peak} [m^{-2} per watt launched]. The requested peak irradiance on the receive plane is

$$\mathscr{I}_{\text{peak}} = P_{\text{atm}} I_{\text{peak}}. \tag{6.4}$$

Definitions. $\mathscr{I}_{\text{peak}}$ is peak beam-normal irradiance [W m^{-2}]; I_{peak} is normalized peak intensity from M5 [m^{-2}].

If $\mathscr{I}_{\text{peak}} > \mathscr{I}_{\text{max}}$, the transmitter is aborted at that epoch. The maximum $\mathscr{I}_{\text{max}}$ is an optional engineering/safety constraint [W m^{-2}]; the M6 reference optical and microwave configurations leave it unset, so the corresponding safety feasibility question is *not* resolved by their positive delivered-power values.

6.4 The power and energy identities

Every stage in Eq. (6.1a) must satisfy

$$0 \leqslant P_{\text{grid}} \leqslant P_{\text{receiver}} \leqslant P_{\text{accept}} \leqslant P_{\text{capt}} \leqslant P_{\text{atm}} \leqslant P_{\text{ap}} \leqslant P_{\text{beam}} \leqslant P_{\text{dc}} \leqslant P_{\text{PV,tx}} \leqslant P_{\text{PV}}. \tag{6.5}$$

These are exact numerical invariants in the implemented algebra, independent of the detailed orbit and wave model. If a result violates them, a factor has been misapplied or a bookkeeping error has occurred.

When operating and unsaturated, the entire chain reduces to a product

$$P_{\text{grid}} = P_{\text{PV}}\,\eta_{\text{pc}}\eta_{\text{e2b}}\eta_{\text{o}}\eta_{\text{atm}}\eta_{\text{capture}}\eta_{\text{rx}}\eta_{\text{grid}}. \tag{6.6}$$

Why not use the product form alone? Because power limits, hard outages, partial eclipses and receiver saturation are nonlinear and time-dependent. The stage-by-stage form preserves their precise location and makes figures interpretable.

6.5 Time integration: instantaneous watts versus annual megawatt-hours

A graph of $P_{\text{grid}}(t)$ shows instantaneous delivered power, but a feasibility study needs energy over an interval. For a uniform time grid $t_0, \ldots, t_N$ with step Δt, the implementation uses trapezoidal weights

$$w_k = \begin{cases} \Delta t/2, & k = 0 \text{ or } k = N, \\ \Delta t, & 1 \leqslant k \leqslant N-1. \end{cases} \qquad T = \sum_{k=0}^{N} w_k = N\Delta t. \tag{6.7}$$

Definitions. w_k is the represented duration [s] for sample k; T is total simulated duration [s]. A one-sample degenerate window is handled separately in the code.

Delivered electrical energy is

$$E_{\text{grid}} = \sum_{k=0}^{N} w_k P_{\text{grid},k}, \qquad \langle P_{\text{grid}} \rangle = \frac{E_{\text{grid}}}{T}. \tag{6.8}$$

Definitions. E_{grid} is energy delivered [J], and $\langle P_{\text{grid}} \rangle$ is its time-mean rate [W]. Convert joules to MWh by dividing by 3.6×10^9 J/MWh.

The integral averages both operating periods and outages. This is why a satellite delivering 5–6 kW when visible can have an annual mean much smaller than that value.

6.6 Availability, capacity factor and end-to-end efficiency

Availability is the fraction of the time window with no blocking reason:

$$\mathscr{A} = \frac{1}{T} \sum_{k=0}^{N} w_k a_k. \tag{6.9}$$

Definitions. $\mathscr{A} \in [0, 1]$ is total delivery availability; a_k is the M6 binary availability indicator. It is not equal to the instantaneous capture fraction η_{capture}.

The code also reports median, 5th percentile and 95th percentile of the sampled P_{grid} distribution. A median of 0 means at least half the sampled epochs yield no grid power; it is compatible with nonzero mean power.

6.6.1 What the capacity-factor denominator actually means

A rated reference-link power is constructed by simulating the configured chain under a defined reference geometry: mean heliocentric distance, mean Earth range, full Sun, normal collector incidence, receiver at zenith, configured pointing characteristics and receiver limits, with a stated atmosphere reference. Call its result P_{rated}.

$$\text{CF} = \frac{\langle P_{\text{grid}} \rangle}{P_{\text{rated}}} \quad \text{for } P_{\text{rated}} > 0. \tag{6.10}$$

Definitions. CF is capacity factor [dimensionless], while P_{rated} is a model-specific reference-link delivered power [W]. This is *not* PV rated power or a guaranteed equipment nameplate rating.

If the reference point is thermally infeasible or triggers beam abort, $P_{\text{rated}} = 0$ and the capacity factor is returned as undefined, not spuriously zero. Because the reference uses window-mean geometry rather than a maximum, it is possible for some configurations to give CF > 1 without a mathematical contradiction.

6.6.2 End-to-end incident-sunlight efficiency

The entire solar-to-grid energy fraction is

$$\eta_{\text{total}} = \frac{E_{\text{grid}}}{E_{\text{inc}}}, \qquad E_{\text{inc}} = \sum_{k=0}^{N} w_k P_{\text{inc},k}. \tag{6.11}$$

Definitions. E_{inc} [J] is the integrated solar energy *incident on the collector*; η_{total} is a dimensionless full-window energy fraction, including outages, PV conversion, transmitter conversion, atmospheric transmission, finite-beam capture and ground conversion.

This is the most comparable single percentage across otherwise different operating modes *provided* the same boundaries and time window are used. Its denominator is incident sunlight, not exported PV electricity; multiplying every individual efficiency in a single representative epoch is not equal to the full-window ratio when conditions change over time.

6.7 Outage and reason statistics

An *outage* is a maximal consecutive run of samples where $a_k = 0$. Its approximate duration is the sum of the sample weights in that run. Runs touching either end of the simulation are *censored*: the true outage began before or ends after the chosen window. The report computes outage count, mean, median, 95th percentile, and longest duration.

For a reason j, its *flagged fraction* is

$$F_j^{\text{flagged}} = \frac{1}{T} \sum_k w_k \mathbf{1}\{j \text{ is active at } t_k\}. \tag{6.12}$$

Its *primary fraction* uses the same formula but counts only epochs where j is the first priority active blocking reason. Thus

$$\sum_{j \in \text{blocking}} F_j^{\text{primary}} = 1 - \mathscr{A}, \tag{6.13}$$

whereas $\sum_j F_j^{\text{flagged}}$ can exceed $1 - \mathscr{A}$ because multiple causes can overlap.

Definitions. F_j^{flagged} and F_j^{primary} are dimensionless time fractions; j indexes a named reason such as Earth

occultation or below-minimum elevation.

6.7.1 Sampling diagnostics

If a satellite remains near Earth's synodic daily cadence, one-hour samples can repeatedly sample nearly the same local hour angles. Let g be the maximum unsampled angular gap between neighboring observed hour angles after sorting them around the full 2π circle. The code reports

$$N_{\text{phases,eff}} = \frac{2\pi}{g}, \qquad \delta f_{\text{edge}} \sim \frac{g}{2\pi}. \tag{6.14}$$

Definitions. g is an angular gap [rad], $N_{\text{phases,eff}}$ an approximate number of effective distinct diurnal phases, and δf_{edge} an order-of-magnitude fractional resolution per window edge. This is an aliasing *diagnostic*, not a rigorous error bound.

The original 3600 s optical sample yields availability about 0.4221, versus about 0.4416 for the 3896 s incommensurate sample, and a roughly 1.2% delivered-energy difference. Subsequent manuscript results should refine their sampling rather than treating hourly figures as exact.

6.8 Worked baseline: what the current simulator actually predicts

For the shipped single-satellite optical example: $A_c = 10\,000$ m^2, roughly 1 AU heliocentric orbit, $D = 10$ m, $\lambda = 1\,\mu$m, a horizontal receiver of physical radius 1 km, 50% electrical-to-beam conversion and 40% receiver optical-to-electrical conversion. For the same baseline orbit, a separate microwave configuration uses 2.45 GHz, a 1 km array, 10 km radius rectenna, 70% electrical-to-beam and 85% receiver conversion.

Model output (one-year scenario)	Optical	Microwave
Mean onboard PV electricity	3.6706 MW	3.6706 MW
Median capture during access	6.31×10^{-3}	3.88×10^{-7}
Mean grid power (all epochs)	1 751.1 W	0.3145 W
Peak grid power	5 937 W	1.073 W
Annual delivered energy	15.35 MWh	2.76 kWh
Availability (hourly grid)	0.42209	0.42209
Capacity factor (reference-link basis)	0.2970	0.2951
Solar-to-grid energy fraction	1.287×10^{-4}	2.31×10^{-8}

These values are *recorded scenario results*, not independent experimental measurements. Different receiver areas and transmitter technologies make the two examples unequal-infrastructure comparisons. Nevertheless, both reveal that finite receiver capture at interplanetary distances is a severe bottleneck in their selected configurations.

6.8.1 Why a much larger optical aperture can improve the result

In the optical 45-point sweep, aperture diameter D varies over 10, 30, 100 m; receiver radius over 1, 5, 10 km; and orbital semimajor axis over 0.3, 0.5, 0.7, 1.0, 1.2 AU. The highest mean grid power in the tested grid is about 604.5 kW for $a = 0.7$ AU, $D = 100$ m, and $a_r = 10$ km, with median available-epoch capture about 0.963.

This is a *parameter-sweep result*, not a formally optimized or proven feasible spacecraft. In the shipped sweep, the receiver has no Sun-avoidance restriction, and part of the 0.7 AU delivery is associated with

low apparent Sun–satellite separation. At 0.3/0.5 AU, the assumed 400 K PV limit prohibits transmission entirely. Such results are scientifically useful precisely because the reason for failure is explicit.

6.9 Reproducible outputs and what each proves

A successful command produces a resolved configuration, manifest, Parquet time series, summary JSON, figures, report and log. The resolved configuration records all default and overridden assumptions. The manifest records versions, hashes, kernels, time-scale conventions and execution environment. The time series stores per-epoch states, access flags, thermal values, expected capture, stage powers and reason codes. The summary computes its metrics *from the saved time series*, making the numbers traceable. Plots are regenerated from the saved files, not redrawn from unrecorded notebook state.

The validation command verifies analytic orbit limiting cases, planetary states, light-time, occultation, flux/thermal balance, beam capture, power-stage monotonicity and end-to-end reproducibility. The source's latest reviewed M6 and M6b documentation reports 37/37 model validation checks passing; the code's broader test suite is larger and has its own CI records. Passing numerical checks supports code reliability *within the implemented equations*; it does not establish hardware feasibility or validate every empirical input.

6.10 Numerical performance and the optional JAX kernel

The default diffraction capture backend is NumPy/SciPy with validated quadrature. For batches of Airy calculations, an optional JAX path evaluates the same capture expressions with fixed quadrature shapes. Because the JAX version used by the repository lacked suitable accurate direct J_0, J_1 kernels for its full argument range, the code constructs those Bessel values using a periodic trapezoidal quadrature of the Bessel integral representation. This is a numerical acceleration/implementation choice, not additional beam physics. CPU reference cross-checks are required; only the capture kernel may be GPU-accelerated, while the full orbital integration is CPU-bound in the examined implementation.

6.11 M6 research interpretation and what remains outside its scope

Verification / interpretation

Definition of done for M6. A validated, single-satellite, single-receiver configuration runs end-to-end without manual intermediate edits; each state and conversion carries consistent frame/epoch/units; each loss has a single owner; time series and summaries are reproducible; unavailable epochs yield zero delivered power; receiver caps bind correctly; improved aperture and shorter wavelength improve the appropriate baseline capture; and plots and reports can be re-created from stored data.

Assumptions and limits

No constellation feasibility claim yet. M6 assumes one transmitter and one receiving site. It has no network scheduling or receiver sharing, no optimized satellite number/phases, no coherent cross-satellite beam synthesis, no realistic clouds-in-time model, no validated high-power thermal blooming / receiver design, no comprehensive deployed mass or economic model, and no GEO baseline. These tasks are deliberately reserved for M7–M9 or later research.

Cross-milestone equation map and symbol index

A.1 Sequential dependency map

Step	Primary output	Input and governing relation
M1	Sun-centred $\mathbf{r}(t), \mathbf{v}(t)$	Six osculating elements → Kepler equation $M = E - e \sin E$ → Cartesian state.
M2	Realistic J2000/SSB spacecraft states	SPICE Sun/planetary ephemerides + gravitational differential acceleration + radial SRP, integrated with DOP853.
M3	Access, R, τ, el, f_{illum} and pointing	WGS-84 and Earth rotation + $c\tau = \|\mathbf{x}_{\text{site}}(t_e + \tau) - \mathbf{x}_s(t_e)\|$ + ray occultation.
M4	P_{PV}, Q_p, Q_w, radiator area and feasibility	$P_{\text{inc}} = SA_cC_{\text{inc}}f_{\text{illum}}$; PV temperature root; two radiative heat balances.
M5	η_{capture} and η_{atm}	Normalized aperture field → diffraction $I(\rho)$ → finite receiver integral → pointing/wavefront/array loss.
M6	$P_{\text{grid}}(t_k)$, E_{grid}, availability, CF	Sequential on-board and ground energy chain with operating policies, followed by weighted time integration.

A.2 Frequently confused variables and names

Symbol / term	Meaning and the pitfall to avoid
a (M1–M2)	Sun-centred orbital semimajor axis [m], not receiver radius or transmitter aperture radius.
$a_{\text{ap}} = D/2$ (M5)	Transmitter aperture radius [m], not orbital semimajor axis.
a_r (M5)	Receiver physical radius [m]; $a_{\text{eff}} = a_r\sqrt{\sin \text{el}}$ for a horizontal disk under the equal-area approximation.
e vs $e_\oplus$	Orbital eccentricity versus WGS-84 ellipsoid first eccentricity.
$\epsilon_f, \epsilon_b, \epsilon_r$	Longwave thermal emissivities, unrelated to orbital e or float64 machine epsilon.
η_{PV} vs η_{capture}	Fraction of *absorbed* sunlight converted to PV electricity versus fraction of *beam-plane power* intercepted by receiver.
β_{SRP} vs $\boldsymbol{\beta}$	Outward radiation acceleration relative to solar gravity versus velocity measured in units of c.
ν vs u	True anomaly from periapsis versus argument of latitude from ascending node.

Symbol / term	Meaning and the pitfall to avoid
σ_{SB} vs σ_θ vs σ_w	Stefan–Boltzmann constant, angular pointing jitter, and RMS wavefront path error, respectively.
ρ vs $\boldsymbol{\rho}$	Scalar transverse beam-plane radius versus vector spacecraft-to-receiver ray separation, distinguished by context and boldface.
R vs $r_\odot$	Earth-link range versus heliocentric solar distance. Their ratio and time-dependence affect different parts of the physics.
N_F	Fresnel number $D^2/(\lambda R)$; a far-field validity indicator, not a received-power efficiency.
T_p vs T_r vs T_{orb}	PV temperature [K], radiator temperature [K], and orbital period [s].
Q_w vs Q_p	Waste heat in the on-board transmitter/electronics radiator versus heat radiated locally by the PV panel.
CF vs η_{total}	Capacity factor is mean grid power / reference grid rating; solar-to-grid efficiency is grid energy / incident solar energy.
η_{point}	Relative expected capture / perfect-pointing capture; may be above 1 in special ring-pattern geometries. It is not the same as η_{capture}.
TDB vs UTC	Dynamical time used internally for orbital propagation versus civil clock representation of configuration epochs.
ECLIPJ2000 vs J2000	Two fixed inertial axis systems related by obliquity rotation; transforming axis directions does not translate the coordinate origin.

A.3 Validation checklist by mathematical requirement

Chapter	Essential mathematical checks
M1	Two-body energy and angular-momentum constancy; Kepler period; element-state round trip; robust high-e Kepler inversion.
M2	SPICE state consistency; DOP853 convergence; Sun-only L1 versus analytic M1; SRP reduced-μ equivalence; full-model REBOUND cross-check.
M3	WGS-84 coordinate accuracy; Earth rotation; light-time fixed-point convergence; target point-ahead; occultation limiting geometry; diurnal aliasing diagnostic.
M4	$S \propto r^{-2}$; efficiency clipping; PV root uniqueness/energy residual; radiator scaling; whole-spacecraft energy ledger closure.
M5	Aperture unit normalization; Airy first null and encircled energy; finite-disk convergence; Rice pointing quadrature; independent HCIPy/ITU reference checks.
M6	Stage monotonicity and exact shutdown zeros; receiver caps; output time integration; reproducibility from config and saved Parquet; provenance audit.

Sources and reproducibility

B.1 Repository snapshot and interpretation policy

This textbook uses the `helioswarm-m0-m6` branch of `extremefattypunch/Project-Dyson-Sphere` as inspected at commit `560675fdf7caa041a14e3d07c32815a4d07ed1e5` on 13 November 2026. If the branch is modified later, its numerical examples, tested assumptions and source line numbers can change. The original implementation plan is *Heliocentric Dyson Swarm Coding Agent Implementation Plan*, version 1.0, 11 November 2026, 28 pages (also committed in the repository's `gpt-insights/` directory).

B.1.1 Primary repository documentation

1. Model assumptions (M0–M6): equation definitions, implementation-specific differences, limitations and references.
2. Milestone reports M1–M6 and M6b: model decisions, validation evidence, results, review findings and fixes.
3. Validation report: recorded tolerance, numerical agreement, external comparisons, and explicit unresolved issues.
4. Compliance matrix: which original PDF requirements are completed, deferred or not applicable.
5. Experiment log: E1–E7 physical scenarios, reproducibility paths and interpretation cautions.
6. Implementation source tree: modules referenced at the start of each chapter.

B.2 Key theoretical and standards sources named in the repository

These are references cited by the code and documentation, rather than a claim that every external publication was independently re-read for this textbook.

Field	Reference named in project documentation
Astrodynamics	D. A. Vallado, *Fundamentals of Astrodynamics and Applications*, 4th ed.; H. D. Curtis, *Orbital Mechanics for Engineering Students*; C. D. Murray and S. F. Dermott, *Solar System Dynamics*.
Orbital ephemerides	JPL DE440 / Park et al. (2021), *Astronomical Journal* 161, 105; NAIF SPICE *Time*, *SPK* and *PCK* Required Reading; NAIF kernel manifests.
Numerical integration	E. Hairer, S. P. Nørsett and G. Wanner, *Solving Ordinary Differential Equations I*, adaptive Runge–Kutta methods.
Geodesy and Earth rotation	WGS-84 (NGA.STND.0036); IERS Conventions (2010), Technical Note 36; PCK IAU orientation conventions.
Radiation and solar constants	IAU 2012 Resolution B2 (au); IAU 2015 Resolution B3 (nominal solar quantities); CODATA 2018 σ_{SB}; Kopp and Lean (2011) solar irradiance.
Wave optics	J. W. Goodman, *Introduction to Fourier Optics*, 3rd ed.; M. Born and E. Wolf, *Principles of Optics*; A. E. Siegman, *Lasers*; Balanis, *Antenna Theory*.

Field	Reference named in project documentation
Atmosphere	Kasten and Young (1989) optical airmass; ITU-R Recommendations P.618, P.676 and P.840, accessed via the versioned `itur` model path in the repository.

B.3 What can and cannot be concluded from the implemented mathematics

A validated physics simulator can support *conditional* statements: *given* an orbital model, an aperture and wavelength, a defined ground receiver, a PV model and an atmosphere assumption, what grid power results? It cannot establish that a specific industrial spacecraft, eye-safe optical ground facility, weather network, or multi-satellite Dyson swarm can be built at acceptable mass or cost. Equally, a low-power result under one technology choice is not a theorem that every heliocentric architecture fails. The correct scientific output is a quantified map of which physical and engineering assumptions lead to usable power and which do not.

The single equation that explains the whole M1–M6 project

At each epoch where operating constraints permit a link, the receiver's grid power is obtained by combining (i) *solar irradiance and collector conversion*, (ii) *the onboard electrical/beam chain*, (iii) *finite-aperture beam capture and atmosphere*, and (iv) *receiver and grid conversion.* Orbital and Earth geometry determine the time-dependent variables and whether each link is available. The mean delivered power is not the largest instantaneous power but the integral of $P_{\mathrm{grid}}(t)$ including all zeros, divided by the duration of the simulated window.